

Reviews or Ratings: Quantifying Information Loss from Coarsening

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This paper quantifies the amount of information that is lost from coarsening signals, in a rate of learning sense. I show how a platform seeking to learn an unknown state should trade off between the informativeness and frequency of submission of reviews (signals). As is the case in many online platforms, I model signals as the realized utility of past consumers, where their average utility is dependent on the unknown quality of a product (state). I study how taste heterogeneity for the product impacts the performance of these systems, especially binary review systems. Information loss from coarsening is increasing in the homogeneity of the population (in terms of their idiosyncratic taste shocks). Moreover, optimal asymmetry of review systems follows the same pattern: as homogeneity increases the platform increasingly isolates extreme signals on one side of the distribution. Regardless of the degree of homogeneity, when the optimal threshold is used a binary review is preferred to a full signal if it is at least 3.25 times as frequent. Finally, I illustrate how information loss can be visualized in posterior space.

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1 Introduction

Transmitting information is costly. Yet, in many instances economic agents rely on other agents freely providing their information about some underlying uncertainty. In many settings, it is deemed infeasible or inappropriate to compensate individuals for sharing their information. In some cases this is due to concerns that compensation may bias the information that agents provide (e.g., Amazon prohibiting vendors from paying consumers for reviews). In other cases, compensation is simply not a viable option (e.g., when eliciting students’ course reviews). As a result, the collector of information is reliant on the goodwill of agents providing their own information. Due to this tension—a principal wants information, but lacks a way to directly incentivize agents to provide it—in many settings platforms attempt to reduce barriers to information sharing as much as possible.

One natural way to reduce frictions in information-sharing is to ask for less information. Intuitively, reducing the number of questions that a reviewer must answer increases the likelihood that they will complete the review. This method of reducing frictions is not without a major drawback, however: the shorter review will elicit and thus provide less information. At the same time, if reviewers are more willing to provide coarser information, then this trade-off is not trivial.

This paper’s main contribution is explicitly modelling and quantifying this trade-off. I show when a platform prefers to elicit coarsened information, and what types of coarsened information it chooses to elicit. I propose a model where eliciting more information from reviewers makes them less likely to submit their signal. A principal (platform), aiming collect information about an unknown state, must trade off between more frequent and better information. When the number of potential reviewers is large it becomes possible to quantify explicitly the trade-off. In this case, I show that the principal’s preference over different review systems is independent of its decision problem, but is heavily dependent on the distribution of reviewers’ signals. The former implies that the platform can apply the same methods across products and settings without having to closely monitor other aspects of the environment. The latter means that there is a strong incentive for the platform to understand how the distribution of signals affects the trade-off between frequency and quality. The model succinctly provides conditions under which coarsened reviews lead to mini-

mal loss of information, and then uses these conditions to justify why certain types of review systems are prevalent. In particular, my results predict the optimality of positively-skewed and the prevalence of binary ratings systems.

In the model, a platform elicits information from a pool of agents (reviewers) who each have a signal that is informative of the underlying state of the world. These reviewers are interpreted as those who have previous experience about the state, and their signal represents their realized utility from this experience. The platform chooses a mapping from signals into reviews. Reviewers observe the mapping, and each chooses whether to submit their signal to the mapping or not. This choice depends on a private willingness to review and the number of possible reviews they can leave: reviewers are less likely to leave a submission when they must decide between many possible reviews.

Motivated by online reviews, I focus on the case when there is a large number of reviewers. As the number of reviewers grows, the state will eventually be known with near certainty as long as a nonzero proportion leave reviews. What matters, then, is the *rate* at which the state is learned by the platform. This rate depends on both the average informativeness of a single review and the probability that a reviewer decides to submit one. Since this rate is independent of the platform’s decision problem, this setting admits a decision problem-free ranking of review systems.

Technically, I leverage large deviations techniques in order to characterize this rate, for each review system. These techniques allow me to characterize the trade-off between frequency and quality of a review. The developed ranking is separable across this trade-off, enabling independent analysis. Since the rate of review can easily be computed empirically, I focus on characterizing the informativeness of a single review for different review systems. A key characteristic of online review platforms is that a single review is almost uninformative: individual taste differences mean that, on average, any one reviewer’s experience is a very noisy signal about the underlying quality of the product. In this context, I derive a straightforward quantification for how much information is lost in terms of how much more frequently the coarse review to be submitted relative to a full-signal review for the platform to prefer the coarse review.

After developing this classification, I study review systems with a finite number of

bins. In practice, many discrete rating systems often degenerate to a review system resembling a binary system. It has been well-documented (see Hu, Zhang, and Pavlou (2009)) that many 5-star review systems degenerate to a bimodal distribution. Even in this simple setting, the platform must determine how much information is lost when a simple positive/negative review is used as well as determine what is the optimal definition of “negative”. I find that the answers to both of these questions are heavily reliant on the dispersion of taste shocks in the reviewers’ signals. Here, larger dispersion reflects more heterogeneous experiences within the population for the product.

A key insight is that a more homogeneous population of reviewers implies that even when information is scarce, some reviewers have very informative signals. When coarse reviews are used, these very informative signals are mixed with a much larger mass of uninformative signals, drowning out their informational content. On the other hand, when reviewers are very heterogeneous, the variance of signal informativeness is small so coarsening does not mix as different information. As a result, more information is lost due to coarsening when the population is homogeneous.

Another key insight of the model is that the optimal threshold for a negative review is extremely important for minimizing information-loss. When reviewers are sufficiently heterogeneous, the optimal binary review system is simple: the platform optimally asks reviewers to symmetrically distinguish between a positive and a negative experience. However, this symmetric review is *not* optimal when reviewers are homogeneous: as agents become increasingly homogeneous, the optimal binary review system becomes increasingly asymmetric. If the platform were to use a symmetric system for homogeneous populations, the amount of information that it would lose would be unbounded. However, the optimal threshold bounds information loss. For any degree of homogeneity, if a reviewer is 3.25 times as likely to submit a binary review as their full signal, the optimal binary system outperforms the full system.

These results suggest that even extremely coarse reviews can be constructed such that they retain a relatively large amount of information. The framework also helps explain why the ratings systems that are observed in practice are implemented, even when they seem counter-intuitive. The stark asymmetry in review systems used by platforms may indeed be optimal, despite their appearance of sub-optimality. For

instance, reviewers of many platforms have converged to skewing their review to be very positive (e.g., many users default to 5-stars on Uber and Airbnb), unless their experience was very negative. I show that if reviewers exhibit less negative than positive dispersion (i.e., they agree more on what constitutes a negative experience than a positive one), it is always optimal for the platform to isolate large negative experiences, while grouping all other experiences together. This results in a much larger number of positive reviews than negative reviews, even when the state is unfavourable.

To further illustrate the intuition behind the above results, I conclude with a discussion on how information loss can be visualized in posterior space. In particular, I show how the informational content of a review can be viewed as the expectation in posterior space of a concave function, and how coarsenings act as pre-evaluation of the function on a partition of posterior space. Hence, information loss can be viewed concretely as arising from Jensen’s inequality applied to a specific family of functions.

The outline of the paper is as follows. Section 2 presents and discusses the model. Section 3 formalizes the trade-off between review informativeness and reviewers’ frequency of review. Section 4 focuses on heterogeneity in the case of binary reviews. Section 5 shows how to visualize information and information loss in posterior space. Section 6 discusses several extensions (including extending results to arbitrary finite review systems) and Section 7 concludes. All proofs are included in the Appendix.

1.1 Related Literature

This paper is closely related to the literature studying learning dynamics in recommendation and review systems.

A large strand of this literature focuses on social learning dynamics. Ifrach et al. (2019) studies a standard social learning setting with binary reviews. Besbes and Scarsini (2018) looks at a similar setting, and focuses on when boundedly-rational agents use of summary statistics of reviews will lead to eventual learning of the state. Most similar to my work is Acemoglu et al. (2022), which studies learning rates for different review systems in the context of social learning. Although both their study and this work are interested in learning rates, there are several major differences. In particular, instead of focusing on the effect of social learning, I focus on quantifying the trade-off between different review systems. Additionally, I focus on a setting where a key difference in different review systems is the rate at which reviews are left.

In the setting of Acemoglu et al. (2022), where reviews are automatically submitted, a finer review system is always preferred to a coarser system; in my setting this is not the case.

There is also a related literature that takes a different approach to studying recommendation and review systems. For instance, Che and Hörner (2018) studies a platform which aims to understand the quality of a product by “pushing” it to consumers. The signal structure in that model is perfect good news, but the platform cannot prove the good news to consumers. Another, very different approach is taken in Garg and Johari (2019), which uses large deviations techniques to optimally differentiate many underlying states by controlling the rate at which positive reviews are given for each quality level. A major difference between this work and Garg and Johari (2019) is that in that model, the designer has full control over the information system, whereas in my case the designer is constrained by the signal structure of the reviewers.

My work is also informed by the empirical literature on ratings. Looking at the common five-star system, it has been well documented that, across platforms, the distribution of submitted ratings is positively-skewed and bimodal (see Chen, Yoon, and Wu (2004) and Hu, Zhang, and Pavlou (2009)). Hu, Pavlou, and Zhang (2017) studies how this distribution is due to a self-selection effect in the population. Fradkin, Grewal, and Holtz (2021) and Fradkin and Holtz (2023) study possible interventions to improve the rate and quality of review submission. These works find that while these interventions (including incentivizing reviews) can increase the rate at which reviews are submitted, but do not tend to improve the overall informativeness of the review system. For a survey of empirical work aiming to understand review systems, see Magnani (2020). Recent empirical work has also provided alternative reasons to use coarsened information. For instance, Botelho et al. (2025) recently showed that moving from a five-star system to a binary rating decreased racial discrimination in an online platform that matches workers with customers. My work compliments this literature by providing an information-theoretic argument for using simple review systems.

My paper contributes also to the literature studying and applying rates of learning to understand economic contexts. This literature is broad, as learning rates have use

in many settings. For instance, Rosenberg and Vieille (2019) and Hann-Caruthers, Martynov, and Tamuz (2018) study how quickly actions converge in a social learning setting, and Harel et al. (2021) and Dasaratha and He (2024) study the impact of networks on social learning.

Technically, this paper applies large deviations techniques to study rates of learning. There is a developed statistical literature that uses similar techniques to compare statistical experiments. Chernoff (1952) provides foundational results in the context of hypothesis testing. Torgersen (1981), which develops a generic ordering over different statistical experiments for finite decision problems when the number of signals is large, extends the ordering of Blackwell (1951). Large deviations techniques have been applied in several cases in the economics literature. Moscarini and Smith (2002) studies more sensitively the rate of learning, in order to determine characteristics of the demand for information. Frick, Iijima, and Ishii (2024b) ranks different mis-specifications by the rate at which they slow learning.¹ Mu et al. (2021) develops a generalization of the ordering of Blackwell (1951) in the context of many signals. Finally, Fedorov, Mannino, and Zhang (2009) looks at similar trade-offs to those that I study in the context of hypothesis testing.

2 Model

There are two states of the world, $\Theta = \{L, H\}$. The state θ is drawn at the start and does not change. There is commonly held prior belief q on the state being $\theta = L$.

There are two classes of agents. The first agent is the principal, a “platform”, which seeks to take an action depending on the state. It has finite action set A and state-dependent utility function $u : A \times \Theta \rightarrow \mathbb{R}$.

The second class of agent is a “reviewer”, which has information about the state of the world θ . There are a finite number N of reviewers, indexed by $i \in \{1, \dots, N\}$. Their information is modelled as i.i.d. signals, as follows. There is an underlying measure space $(\Omega, \mathcal{F}, P)$, and a real-valued random variable according to each state S_i for reviewer i such that given θ all of the S_i are independent and identically distributed, with densities given by f_θ (with $f_L \neq f_H$). I assume that the S_i take values in $\mathbb{R}$, so that information is one-dimensional, with realization denoted s_i .

¹Frick, Iijima, and Ishii (2023) and Frick, Iijima, and Ishii (2024a) apply similar techniques to different settings.

In addition to their own private signal, reviewers are endowed with a willingness to review W_i (with realization w_i), another real-valued random variable. I assume that the W_i are i.i.d. and that W_i and S_i are independent. This ensures that the distribution of signals that are submitted across review systems is the same.

The platform is interested in maximizing its expected utility, and achieves this goal by eliciting information from reviewers. Specifically, the principal chooses a review system, which is a (measurable) function $r : \mathbb{R} \rightarrow \mathbb{R}$. If a reviewer submits signal s_i under review system r , the platform does not directly observe s_i but rather observes only $r(s_i)$. Let $R_r = r(\mathbb{R})$ denote the set of reviews possible under r .

Associated with each review system is a “cost” to review, that is borne by the reviewer. This can be interpreted as either a cognitive or temporal cost associated with submitting the review. The cost of a review is given as $c(r) = C(|R_r|)$ for some increasing function $C : \mathbb{N} \cup \{\infty\} \rightarrow \mathbb{R}$. That is, the cost of the review is directly related to the number of different reviews that can be submitted. This captures the intuition that it is “easier” to submit reviews when there are fewer choices.²

After the platform makes its choice of review system, reviewers arrive. The reviewers’ private signals and willingness to review are drawn, and reviewers then decide whether to submit their review. Reviewer i gets intrinsic utility $w_i - c(r)$ from reviewing, and 0 utility from not. It follows that reviewers will use a simple threshold rule: if $w_i \geq c(r)$ they will review, and if $w_i < c(r)$ they will not. Let $p_r := P(W_i \geq c(r))$ be the ex ante probability that a reviewer submits a review r .

After the N reviewers make their review decisions, the platform observes the submitted reviews, update its beliefs, and then takes an action to maximize its expected utility. Explicitly, let I be the set of reviewers who decide to submit a review. The platform observes $\mathcal{S}(I) = (\tilde{r}(s_1), \dots, \tilde{r}(s_N))$, which is the set of *submitted* reviews:

$$\tilde{r}(s_i) = \begin{cases} r(s_i) & \text{if } i \in I \\ \emptyset & \text{if } i \notin I \end{cases}.$$

²Much recent work has shown that individuals have preferences against complexity. See, for instance, Oprea (2020). In some cases, complexity has been conceptualized in a way dependent on the number of objects under consideration (e.g., Puri (2022)). Recently, Wang and Li (2025) studied the impact of increased cognitive load on user engagement in such systems, and finds that decreases in complexity lead to more user engagement.

After observing the set of reviews, the platform updates its belief that the state is L . Let $\pi(\mathcal{S}) = \pi(L|\mathcal{S}(I), q)$ denote the posterior belief on the state being L after observing reviews $\mathcal{S}$ with prior q . Here, and below, I drop the reliance on q and I for notational clarity.

After updating beliefs, the platform then chooses an action to maximize its expected utility given beliefs $\pi(\mathcal{S})$:

$$u^*(\mathcal{S}) := \max_{a \in A} \{ \pi(\mathcal{S})u(a, L) + (1 - \pi(\mathcal{S}))u(a, H). \}$$

Since $\mathcal{S}$ is a random vector, $u^*(\mathcal{S})$ is also necessarily stochastic. Hence, the platform aims to maximize the expected value of this object, which depends on r and N :

$$u^*(r, N) := \mathbb{E}[u^*(\mathcal{S})],$$

where this expectation is taken over the S_i and W_i . The objective of the platform is to maximize $u^*(r, N)$ by optimally choosing r .

2.1 Discussion of Model

This model flexibly captures situations where an agent must use information elicited from others in order to make informed decisions but the agent cannot force or otherwise incentivize the group with information to share their signals.

Importantly, I do not take a stance on what is the goal of the platform. In some settings, it may be the case that the platform’s goal is to share the information that it collects with other agents (e.g., consumers of a product that is of unknown quality). The model naturally nests this setting, but does not assume that the platform’s goals are aligned with these future consumers. In this way, the model sheds insight on a variety of settings, from online retailers to general information dissemination platforms like Rotten Tomatoes.

This highlights a major distinction between the goals of the platform in my model and previous models of review systems. For instance, the platform’s goal in designing a rating/recommendation in Acemoglu et al. (2022) and Che and Hörner (2018) is to maximize the expected utility of consumers. In my work, the platform’s incentives

may not be aligned directly with consumers. This distinction in focus is due to the primary focus in those works being how platforms can incentivize early consumers to explore a product of unknown quality, whereas I abstract from the problem of experimentation. For this reason, I view the present work as complimentary with the aforementioned studies. Previous works provide guidance on how a platform should design the information that it provides to early consumers to incentivize experimentation, while my study suggests how the platform should collect information in order to further its own goals. In practice, it should aim to maximize the information that it collects for itself, and then consider what information is optimal to show to consumers; the two objectives are not incompatible.

One of the main restrictions in the set-up as outlined above is the assumption that, given a review system r , the reporting rate p_r is uniform across the population of reviewers. In practice, this is a restrictive assumption: it is natural that reviewers with more extreme signals are more likely to submit reviews.³ To account for this, I extend the model to allow for non-uniform reporting rates in Section 6.1. The main insights of the baseline model extend to that setting, and so I begin with a discussion of the uniform reporting rates for clarity of exposition.

As a final note of discussion on the model, I assume throughout that there are no processing costs for the platform to interpret the submitted reviews. One of the practical benefits for using simple rating schemes is that they are easier to digest and summarize than full-length reviews. My model abstracts from this, and assumes that there is no cost to interpreting the reviews. Closely tied to this is the possibility of misinterpretation or miscommunication that are inherent to complicated reviews (i.e., free-text). I assume that when a reviewer submits a review, that it is reported perfectly. Abstracting from both of these forces suggests that my model overstates the amount of information loss that stems from using coarsened signals, since removing these forces makes full-signal reports more accurate.

3 Quantifying Information Loss

In this section, I quantify the trade-off between informational content and frequency of report for different review systems by applying large-deviations techniques. This

³See, for instance Lafky (2014).

quantification admits a generically complete ranking of review systems that is independent of the platform's decision problem.

Fix a review system r . Denote the measure of reviews under r conditional on state L as γ_L^r . That is, for any measurable $B \subseteq R_r$ (where measurability is inherited from the Borel σ -algebra),

$$\gamma_L^r(B) := \mathbb{P}_L(r^{-1}(B)) = \int_{\{s:r(s) \in B\}} f_L(s) ds.$$

Similarly, let γ_H^r denote the measure corresponding to the distribution of reviews in state H . These measures reflect the distribution of reviews r *conditional* on reviews being submitted.

As mentioned, the performance of a review system depends on the rate at which the state is revealed under the review system. Intuitively, the more different are γ_H^r and γ_L^r , the faster will be learning. The measure of “different” that is correct in this context is a standard notion of divergence, the Hellinger transform between the two distributions.

Definition 1. Define

$$\nu(r) := 1 - \min_{\lambda \in [0,1]} \int_{R_r} \left(\frac{d\gamma_L^r}{d\gamma_H^r} \right)^\lambda d\gamma_H^r$$

to be the *learning efficiency* of review system r .

The learning efficiency is a measure of divergence between the distributions γ_L^r and γ_H^r . The more different the distributions in the two states, the less overlap between the two distributions, and the larger the learning efficiency becomes. At one extreme, when $d\gamma_L^r > 0 \iff d\gamma_H^r = 0$, then $\nu(r) = 1$, so that learning occurs after one review is received. At the other extreme, when signals are entirely uninformative, $\gamma_L^r = \gamma_H^r$, and so $\nu(r) = 0$, and so learning never occurs.

The reason that the learning efficiencies $\nu(r)$ are important is that when all reviews are submitted (i.e., $p_r \equiv 1$), these $\nu(r)$ capture the rate (in terms of number of reviewers N) that the platform will learn the state. In particular, this means that, when N is large, the probability that the platform takes an action that is not optimal given the state is determined by $\nu(r)$. This means that, *regardless* of the decision

problem, when N is large, $\nu(r)$ determine how quickly the expected utility of the platform converges to the full-information utility. This means that, if $\nu(r) > \nu(r')$, if all reviews are submitted, then for large N the platform will have higher expected utility under r than under r' .⁴

An additional wrinkle in this setting is that signals are not always submitted. Since reviewers may choose not to submit their signal, the rate at which the platform comes to learn the state is no longer $\nu(r)$. The lower the probability that a reviewer reports a signal, the slower the rate. The main result of this section characterizes that the learning rate depends exactly proportionally on p_r . This means, regardless of the decision problem, for large enough N it is $p_r \nu(r)$ that determines the welfare of the platform. This suggests that while the platform must trade-off between the rate at which reviews are submitted and the quality of the reviews, the trade-off is easily separable.

Theorem 1. *Fix two review systems r, r' with corresponding rates of review $p_r, p_{r'}$ such that $p_r \nu(r) > p_{r'} \nu(r')$. Fixing a decision problem for the platform, there exists an $\bar{N}$ such that for all $N \geq \bar{N}$, such that $u^*(r, N) > u^*(r', N)$.*

Theorem 1 provides a straightforward way for a platform to trade-off between frequency of different reviews and their quality. As mentioned, this learning efficiency $p_r \nu(r)$ is independent of the decision problem of the platform. This means that the platform need not tailor its review design on the specifics of each problem but can instead apply the same insights to different settings.

In general, the p_r are easy to empirically compute for platforms. For instance, a large platform can easily implement an A/B test to determine the relative reporting rates for two different review systems. This allows me to focus directly on the computation and comparison of the $\nu(r)$. Another way to view Theorem 1 is that, if

$$\frac{\nu(r')}{\nu(r)} < \frac{p_r}{p_{r'}}, \quad (1)$$

then there exists an $\bar{N}$ such that for all $N \geq \bar{N}$, such that $u^*(r, N) > u^*(r', N)$. Hence, computing $\frac{\nu(r')}{\nu(r)}$ determines the benchmark (in terms of relative frequency

⁴This discussion summarizes a known result in the literature (see, for instance, Theorem 3.9 in Torgersen (1981)).

of reporting) that the platform should use in choosing between two review systems. Moreover, by using the same baseline r' for each r gives a single number for each r . A natural choice for this baseline comparison is the full-review system. Denote by ν the learning efficiency of the review system $r = \text{id}$.

Definition 2. Define the *learning loss* of a review function r as $\kappa(r) := \frac{\nu}{\nu(r)}$.

The learning loss $\kappa(r)$ of a review system r is how much more frequently a review of type r needs to be submitted relative to a full review for the platform to prefer using reviews of type r to full reviews. To compare two different review systems r and r' , the platform simply uses $\kappa(r)/\kappa(r')$. Of course, for any review system, learning loss is at least one. The main work of the following sections is to derive properties of the learning loss in different settings and for different review systems.

Remark 1. It is important for the separability of $\nu(r)$ and p_r that the reporting rate p_r is uniform across the population of reviewers. If this is not the case (i.e., individuals with more extreme signals are more likely to leave a review), then the two are no longer completely separable. However, insights from the general model still apply in that setting. See Section 6.1 for an extension that allows an individual reviewer's reporting rate to depend on their realized signal.

Remark 2. One must be careful not to interpret these ν 's as the “strength” of the signal. It is not that the platform is trading off identically between p_r signals of type r and $p_{r'}$ signals of type r' . The uncertainty in the generation of signals is important for this specific classification. An alternative to this approach is to ask how many signals from r “equals” a signal from r' . I discuss this alternative and its relationship to my specification in Section 6.3.

4 Taste Heterogeneity and Binary Systems

The results of Section 3 do not rely on any specific assumptions on the distribution of signals. Indeed, it is not necessary to assume that signals have real-valued or have density. In this section I restrict attention to interpreting the state as the underlying quality of a good/product, reviewers as previous consumers of the good, and their signals as their realized utility of consuming the product.

Explicitly, I assume that the reviewer's signal is given by $s_i = u_\theta + x_i$, where

$$u_\theta = \begin{cases} -\mu & \text{if } \theta = L \\ \mu & \text{if } \theta = H \end{cases}$$

(for $\mu \in \mathbb{R}_+$) is the baseline utility that the reviewer gets from the product, and x_i is reviewer i 's real-valued idiosyncratic taste shock. The normalization of $u_\theta = \pm\mu$ is simply to maintain symmetry, and is without loss. Let μ be called the *degree of distinction*, as it determines how much the distribution of signals in L differs from the distribution of signals in H . For large μ the two look very different, and for small μ the two are almost indistinguishable.

Assumption 1. Let X_i be the random variable reflecting reviewer i 's taste heterogeneity. The taste heterogeneities X_i are i.i.d., with continuous, full-support, log-concave density given by f .

With these assumptions on the structure of the reviewers' signals, the density of signals in state L is given by

$$f_L(s) = f(s + \mu),$$

and that the density in state H is given by

$$f_H(s) = f(s - \mu).$$

It is important to note that writing the signals in this way assumes that the taste shocks are independent of the underlying quality of the good.

The two primitives at this stage are f and μ . Given a review system r , write the learning efficiency of r when the degree of distinction is μ as $\nu(\mu; r)$, dropping the implicit dependence on f . Similarly, write the learning efficiency of the full review ($r = \text{id}$) as $\nu(\mu)$, and the learning loss of r as $\kappa(\mu; r)$.

4.1 Binary Review Systems

In this section I focus on the simplest non-degenerate coarsenings: binary systems. In practice, many ratings systems are binary (e.g., likes/dislikes on YouTube, "would or

would not recommend” questions on many platforms, up-/down-votes on Reddit), or are close to binary in that the distribution of ratings is very bimodal (see Hu, Zhang, and Pavlou (2009)), and so this is a natural setting to study. Specifically, suppose that the platform is choosing between eliciting a full review from the reviewers or asking for a specific binary review. The platform is able to choose any binary review. So, the comparison that the platform is making is between the full signal and the *optimally chosen* binary review. The qualitative insights remain the same when the platform is choosing across finite review systems with $b > 2$ bins, so I focus here on the binary case for clarity of exposition (see Section 6.2 for a discussion of arbitrary finite systems).

In the context of online reviews, individual signals are typically very uninformative. Hence, the most applicable setting is that in which the two distributions f_L and f_H are very similar. This corresponds exactly to the case that $\mu \approx 0$. For any review system r (including the full review), $\lim_{\mu \rightarrow 0} \nu(\mu; r) = 0$. However, learning loss remains well-defined in the limit, and so I focus on

$$\lim_{\mu \rightarrow 0} \kappa(\mu; r).$$

First, because f is log-concave, it will always be the case that an optimal binary system r will be defined by a single threshold τ :

$$r(s) = r_- \cdot \mathbb{1}\{s \in (-\infty, \tau]\} + r_+ \cdot \mathbb{1}\{s \in (\tau, \infty)\}.$$

This is because, f being log-concave guarantees that the more negative the signal, the larger the posterior of the state being $\theta = L$. It is intuitive then that a threshold rule will be optimal in posterior space: signals that agree more on the underlying state should be grouped.

Lemma 1. *Suppose that f is log-concave. Then for any μ , consider a review system r' with $|R_{r'}| = 2$. There exists a b -threshold rule r such that $\nu(\mu; r) \geq \nu(\mu; r')$.*

This means that the platform must optimize over the threshold τ : it chooses a bar for reviewers, who then report whether their realized utility was above or below this bar. For notational clarity, I abuse notation and write τ for the review system

defined by threshold τ . That is, I write $\nu(\mu; \tau)$ and $\kappa(\mu; \tau)$.⁵ Summarizing, the object of interest is, for a fixed f and a given τ ,

$$\lim_{\mu \rightarrow 0} \frac{\nu(\mu)}{\nu(\mu; \tau)} = \lim_{\mu \rightarrow 0} \kappa(\mu; \tau). \quad (2)$$

Recall that this limit represents a threshold in the relative rates of reporting that make the platform prefer eliciting binary reviews to eliciting full reviews. If the limit (2) is large, then when reviews are not very informative, a binary review system with threshold τ performs poorly relative to full reviews. On the other hand, if (2) is small, then even when reviewers are slightly more likely to leave a binary review, then the platform would prefer asking for the coarse information.

The main result of this section is that, the limit in (2) has a tractable form that can easily be interpreted. Before I introduce the result, some intuition is useful. To this end, consider the numerator and denominator of the limit separately. When $\mu = 0$, $\nu(\mu) = 0$, since there is no difference in the distribution of signals in the two states. When the degree of distinction changes from being zero to positive, consider the forces that most lead to an increase in $\nu(\mu)$. On regions where f is nearly constant, signals will continue to be uninformative. However, on regions where f is very curved, even a small shift in μ causes signals in that region to become very informative, because the posteriors that they induce will shift greatly. It is not surprising, then, that a larger curvature of f is associated with a larger value of (2). The measure of curvature that turns out to be correct is the commonly used *Fisher Information* of f (with respect to μ):

$$I_f := \mathbb{E} \left[\left(\frac{f'(s)}{f(s)} \right)^2 \right].$$

A similar object determines the rate at which $\nu(\mu; \tau)$ increases as μ increases from zero. Since this review is discrete, the effect of a change in μ will only be felt at the threshold τ . Loosely, μ increases, an amount of mass $f(\tau)$ shifts only in state L to the lower bin, and an amount $f(\tau)$ shifts only in state H to the higher bin. This causes imbalance in the two bins, which is what allows for the distinction of the two

⁵Of course, optimally τ will depend on μ . However, in the case of $\mu \rightarrow 0$, it is sufficient to set $\tau^* = \lim_{\mu \rightarrow 0} \tau^*(\mu)$ as $\lim_{\mu \rightarrow 0} \frac{\nu(\mu; \tau^*(\mu))}{\nu(\mu; \tau^*)} = 1$.

states. Hence, the larger $f(\tau)$, the more of an affect a shift in μ will have. This effect is magnified if the mass in one of the bins is small because the marginal mass $f(\tau)$ will then have a greater effect for that bin.

The following assumption is just a regularity assumption on f , guaranteeing that the Fisher information is well-defined.

Assumption 2. Assume that f is twice continuously differentiable almost everywhere, with the first and second derivative of $f(x + \mu)^\lambda f(x - \mu)^{1-\lambda}$ with respect to μ integrable for μ in a neighbourhood of 0.

These effects come together in the following major result of this section:

Theorem 2. Suppose that f that satisfies Assumptions 1 and 2. Let F denote the CDF of f . Then,

$$\lim_{\mu \rightarrow 0} \kappa(\mu; \tau) = I_f \cdot \frac{(1 - F(\tau))F(\tau)}{f(\tau)^2}.$$

Notice that this term must be weakly larger than 1, since for any positive degree of distinction, the binary review necessarily loses information. In addition to the intuition from above, the result of Theorem 2 shows that, given a distribution of taste shocks, it is straightforward for the platform to determine what is the optimal threshold:

Corollary 1. The optimal binary threshold given a distribution is given by the value of τ that maximizes

$$\frac{f(\tau)^2}{F(\tau)(1 - F(\tau))}.$$

Given the importance of the distribution for this term, it is clear that the particulars of f are integral for determining what is the optimal threshold τ . The next section investigates one such property of f .

Remark 3. Instead of taking the limit of $\mu \rightarrow 0$, one could just as naturally consider the impact of increasing the variance of taste shocks. That is, fix μ and instead parameterize the distribution by its variance σ :

$$f_L(s; \sigma) \propto f\left(\frac{s + \mu}{\sigma}\right) \quad \text{and} \quad f_H(s; \sigma) \propto f\left(\frac{s - \mu}{\sigma}\right).$$

The equivalent of taking $\mu \rightarrow 0$ with this parameterization is taking $\sigma \rightarrow \infty$. The choice between the two is stylistic: the two are equivalent for my purposes. Indeed, it is not hard to see that a parameterization of (μ, σ) yields the same information as $(\frac{\mu}{\sigma}, 1)$. Hence, the limit of learning loss in the two settings is the same.

While it is natural to keep the baseline utility of the product fixed, a drawback of varying σ is that the notation regarding an optimal threshold becomes more cumbersome. Instead of having a well-defined limit τ , instead the convergence is in terms of $\frac{\tau(\sigma)}{\sigma}$. Hence, I maintain throughout the convention that $\sigma \equiv 1$, and that $\mu \rightarrow 0$. I emphasize again that the two are equivalent.

4.2 Heterogeneity

An important economic characteristic when eliciting information is how heterogeneous or homogeneous is the population of individuals whose information is being elicited. In the context of product reviews, different products are likely to induce a different degree of homogeneity in the population. Preferences over products like movies (or other forms of art) are likely to be quite dispersed because individual taste matters to a large degree. However, for simple products like toasters, preferences are likely to be much less dispersed, since experiences are much more uniform.⁶ Since homogeneity of experience differs so greatly across products, it is important to understand how it affects review systems.

In this section I use the results from the previous sections to show how taste heterogeneity affects both the performance of binary rating systems and the characteristics of optimal binary rating systems. To this end, I restrict attention to a one-parameter family of distributions where the parameter can be interpreted as the degree of homogeneity within the population of reviewers:

$$f_{\alpha}(s) \propto \exp(-|s|^{\alpha}). \quad (3)$$

⁶One possible way to distinguish between goods with low and high homogeneity would be the classification of “search” and “experience” goods. First introduced by Nelson (1970), search goods are those goods whose attributes are well-defined and easily found, whereas experience goods are those which must be experienced before an opinion can be formed (see also Magnani (2020) for a discussion). Search goods, because of their well-defined attributes, are likely to be more homogeneous (does the good perform as it should), while experience goods are likely to be more heterogeneous (since individual experience matters more).

A larger dispersion of taste shocks corresponds to a larger heterogeneity in the population as it corresponds to a wider experience given the state. As discussed in Remark 3, it is not sufficient just to increase the variance of signals. Instead, it is the tail properties of the distribution that impacts learning loss. Larger tails correspond to greater dispersion, and hence greater heterogeneity. The parameter α in the family of (3) captures exactly this tail behaviour. The larger the value of α , the smaller are tails, and so, the smaller the dispersion of taste shocks; the population is more homogeneous.

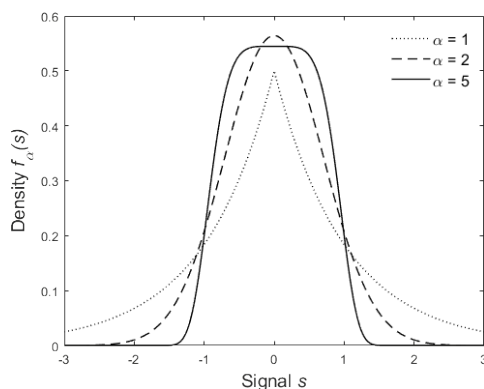


Figure 1: f_α for $\alpha = 1, 2, 5$.

Remark 4. For $\alpha \geq 1$, $f_\alpha(s)$ is log-concave, and so from Lemma 1 a threshold rule will be optimal. This family is sometimes referred to as the Generalized Normal Distributions and incorporates three well-known distributions as special cases: f_1 is the Laplace distribution; f_2 is the Normal distribution; and as $\alpha \rightarrow \infty$, f_α converges to the uniform distribution on $(-1, 1)$. The Laplace distribution corresponds to the largest possible dispersion while retaining log-concavity, the Normal distribution reflects an intermediate level of heterogeneity, and the uniform distribution reflects no heterogeneity. Notice that because variance does not matter, the scale/variance parameter is irrelevant and so is normalized to 1. Figure 1 exhibits $f_\alpha(s)$ for different values of α .

In what follows, let $\nu_\alpha(\mu)$ denote the full review learning efficiency with degree of homogeneity α and degree of distinction μ , and similarly $\nu_\alpha(\mu; \tau)$ the learning

efficiency under the binary with threshold τ . Denote by (with some abuse of notation)

$$\kappa_\alpha(\tau) := \lim_{\mu \rightarrow 0} \frac{\nu_\alpha(\mu)}{\nu_\alpha(\mu; \tau)}$$

the ratio in (2) under this specification, for a given α .

Studying this class of distributions allows me to show how heterogeneity impacts information loss and the optimal choice of threshold. To motivate why properties of the optimal threshold are important, I first show what happens when the platform uses a naive threshold: simply asking reviewers if their experience was positive or negative. This corresponds to the (symmetric) threshold of 0.

Perhaps surprisingly, when the population is very heterogeneous, this threshold leads to small loss in learning. However, as homogeneity grows, this does not continue: the naive threshold leads to unbounded loss in information as homogeneity α increases.

Lemma 2. *Consider learning loss under the naive threshold 0, $\kappa_\alpha(0)$. Then,*

(i) $\kappa_1(0) = 1$; and

(ii) $\kappa_\alpha(0)$ is increasing in $\alpha \geq 1$ with $\lim_{\alpha \rightarrow \infty} \kappa_\alpha(0) = \infty$.

Lemma 2 shows that under the naive binary review, information loss is increasing in the degree of homogeneity of the population. The interpretation is fairly straightforward: when individuals are more homogeneous, there is a larger possibility of individuals having extreme signals that are reflective of the underlying state. This is because, when taste shocks are less dispersed, while many signals are very uninformative, there are also more signals that are very informative (i.e., some reviewers are very representative). In posterior space, this is reflected in a larger spread of the induced posteriors. Since coarsening to a binary review mixes these informative signals with many uninformative signals, much more information is lost. This is explored more in Section 5, where it is shown how to explicitly visualize learning loss in posterior space.

This highlights the central tension in choosing an optimal threshold: extracting more information from some signals while extracting less from others. The solution to the trade-off is not obvious. For values of α that are close to 1, $\kappa_\alpha(0)$ is close to the

lower bound of 1, so it is unlikely that a different threshold could improve upon this. However, as discussed above, when α is large, a small movement in μ induces some extremely informative signals. This suggests that the incentive to choose a threshold that is not 0 is increasing as α grows.

As shown below, this intuition turns out to be correct. As α increases, the optimal threshold becomes increasingly asymmetric. When the threshold is chosen optimally, the result from above that the amount of information that is lost is unbounded can be undone. While it is still true that information loss is increasing in the degree of homogeneity of the population, this information loss is now bounded above. If the population is 3.25 times as likely to submit a binary review than their full signal, the binary system should be preferred, regardless of the degree of homogeneity.

In what follows, let $\tau^*(\alpha)$ be the nonnegative⁷ value that maximizes

$$\frac{f_\alpha^2(\tau)}{F_\alpha(\tau)(1 - F_\alpha(\tau))}. \quad (4)$$

Proposition 1. *For $\alpha \geq 1$, there is a unique maximizer $\tau^*(\alpha)$ to (4) on $[0, \infty)$. It has the following properties:*

- (i) *If $\alpha \leq 2$, $\tau^*(\alpha) = 0$, while if $\alpha > 2$, $\tau^*(\alpha) \neq 0$;*
- (ii) *$\lim_{\alpha \rightarrow \infty} \tau^*(\alpha) = 1$; and*
- (iii) *$\kappa_\alpha(\tau^*(\alpha)) < 2e^2 \int_1^\infty \frac{e^{-s}}{s} ds$ for all α , with equality in the limit. This value is less than 3.25.*

Interestingly, the normal distribution represents the point where the optimal threshold goes from being symmetric to being asymmetric. It is exactly those populations that are more homogeneous than the normal distribution that have a nonzero optimal threshold: When the degree of homogeneity is large the platform optimally asks individuals either (i) whether they had a very good experience or an experience that is not exceptional (positive τ^*), or (ii) whether they had a very bad experience or an experience that wasn't horrendous (negative τ^*).

⁷Since f_α is symmetric for all α , there will be also a nonpositive optimal $-\tau^*(\alpha)$. In the statement of Proposition 1, everything is written in terms of the nonnegative optimal τ^* . Of course, comparable statements for the nonpositive optimal threshold hold.

Figure 2 plots some comparative statics on $\tau^*(\alpha)$: both $\tau^*(\alpha)$ and $F_\alpha(\tau^*(\alpha))$ are increasing in α . These show that not only does the optimal threshold increase in α , the optimal degree of asymmetry in terms of review frequency also increases in α . This means that, for large α , even when the state is high, the majority of reviews will be negative. This fraction increases to 1 in the limit, showing that for large degrees of homogeneity, the optimal review system appears degenerate. These results are summarized in Proposition 2. Figure 2 also shows that, despite the improvements coming from the optimal threshold, learning loss is still increasing in homogeneity.

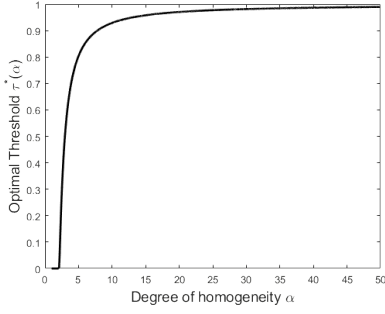
Proposition 2. *The following are true:*

- (i) *The optimal threshold is increasing in the homogeneity of the population: $\tau^*(\alpha)$ is strictly increasing in α for $\alpha \geq 2$; and*
- (ii) *Optimal asymmetry is increasing in the homogeneity of the population: $F_\alpha(\tau^*(\alpha))$ is strictly increasing in α for $\alpha \geq 2$.*

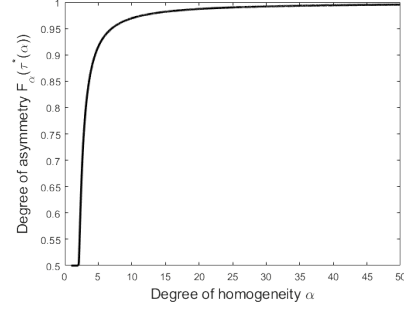
4.3 Asymmetric Heterogeneity

While the results above provide insight about possible optimal review systems, one major drawback is that they suggest that there are two optimal thresholds that result in very different behaviour. If the platform were unable to consistently coordinate reviewers, then the review system could easily break-down. This section shows that the presence of both a positive and a negative optimal threshold disappears when any asymmetry in the distribution of taste shocks is introduced. By asymmetry, I mean different degrees of homogeneity of taste shocks for positive and negative experiences. For any nonzero difference in homogeneity on the two sides of the distribution, the platform optimally chooses a threshold on the side of the distribution that is more homogeneous.

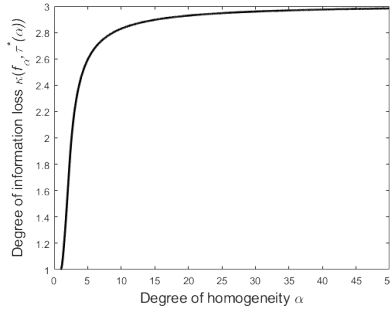
Before stating the result, it is important to highlight why this is a useful setting to study. In many cases, negative experiences are more homogeneous than positive experiences. In hiring a taxi or ride-share service, the qualities that lead to a very bad experience are likely to not differ greatly across individuals: no one wants to be late or get into an accident. On the other hand, the qualities that lead to a very good experience may differ across individuals: some (not all) people like having a



(a) Optimal thresholds as a function of homogeneity.



(b) Optimal degree of asymmetry, as a function of α .



(c) Minimal information loss, as a function of homogeneity.

Figure 2: Plots showing the comparative statics of Proposition 2.

long discussion with their driver, some (not all) like music to be played.⁸ In a context like this, where negative experiences are more homogeneous, the platform optimally uses negative reviews only for very bad experiences and uses positive reviews for any experiences that are at least mediocre. This suggests a robust skewing towards positive reviews. This is a characteristic that is observed in many review systems: most reviews on Airbnb are 5-stars, as are those on Uber (see for instance Hu, Zhang, and Pavlou (2009)).

Let α^* be the value for which $f_\alpha(0)$ is maximized.⁹

Proposition 3. *Let*

$$f_{\alpha_L, \alpha_H}(s) \propto \mathbb{1}\{s \geq 0\}e^{-|s|^{\alpha_H}} + \mathbb{1}\{s < 0\}e^{-|s|^{\alpha_L}}$$

⁸As another example, for most search goods, less negative dispersion is expected: most reviewers will be in agreement about whether it doesn't meet expectations, but there may be disagreement about how it over-performs.

⁹It is easy to verify that $\alpha^* \approx 2.166$.

with $\alpha^* \leq \alpha_H < \alpha_L$. Then, the optimal threshold $\tau^*(\alpha_L, \alpha_H)$ is negative.

Figure 3 shows what is meant by varying degrees of homogeneity, and highlights the intuition behind Proposition 3. Consider using a positive threshold, for instance at the positive dotted line. By looking at the corresponding negative threshold that yields the same value of the density (dashed line), it is easily seen that there is less mass to the left of that threshold than there is to the right of the positive candidate. This logic applies to any such threshold, so that the optimal threshold must be negative. A similar logic applies if, instead of increasing the homogeneity on one side of the distribution, the variance of signals on one side of the distribution were increased. In that case, to, the optimal threshold would necessarily be negative.

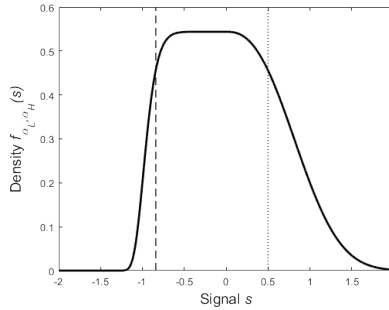


Figure 3: f_{α_L, α_H} , for $\alpha_L = 10, \alpha_H = 2.5$.

As mentioned, Proposition 3 helps provide justification for why in practice platforms like ride-sharing apps having extremely asymmetric systems robustly exhibit many more positive reviews than negative reviews. If people agree on negative experiences more than they do positive experiences, for optimal learning, the platform *should* to isolate extremely negative experiences and thus choose a system that frequently degenerates on a large portion of the population leaving the same review, regardless of the underlying state.

5 Posterior Visualization

In this section I show how the forces that impact the performance of different review systems can be viewed in posterior space. In particular, this visualization provides additional insight to the results of Section 4.

Let $f(s) = \frac{1}{2}(f_L(s) + f_H(s))$ be the ex-ante distribution of signals under a symmetric prior. Moreover, let $\pi(s) := \frac{f_L(s)}{f_L(s) + f_H(s)}$ be the posterior belief placed on state

L after observing signal s when there is a symmetric prior.

A straightforward computation shows that the learning efficiency under the full review is an expectation that depends on the posterior and the ex ante distribution:

$$\nu = 1 - \min_{\lambda \in [0,1]} 2 \int \pi(s)^\lambda (1 - \pi(s))^{1-\lambda} f(s) ds.$$

Associated with each signal s is a unique posterior π . Let γ_π be the measure on $[0, 1]$ that reflects the distribution of posteriors that is associated with f .¹⁰ That is,

$$\gamma_\pi(B) := \frac{1}{2} \int_{\{s: \pi(s) \in B\}} f_L(s) + f_H(s) ds.$$

Then, the above can be rewritten as

$$\nu = 1 - \min_{\lambda \in [0,1]} 2 \int \pi^\lambda (1 - \pi)^{1-\lambda} \gamma_\pi(d\pi). \quad (5)$$

This structure is preserved when review systems are introduced, as long as the review system treats posteriors consistently. That is, as long as it treats two signals that induce the same posterior the same. To this end, call a review system r *consistent* if, for all signals $s, s' \in \mathbb{R}$ such that $\pi(s) = \pi(s')$, then $r(s) = r(s')$.

The structure observed in (5) extends to consistent review systems. The difference is that instead of the function $\pi^\lambda (1 - \pi)^{1-\lambda}$ being evaluated at each π , the review system r groups all posteriors that are mapped to the same review.

Proposition 4. *Let r be any consistent review system. Then,*

$$\nu(r) = 1 - \min_{\lambda \in [0,1]} 2 \int \left(\mathbb{E}_{\gamma_\pi} [\tilde{\pi} | r(\tilde{\pi}) = r(\pi)] \right)^\lambda \left(1 - \mathbb{E}_{\gamma_\pi} [\tilde{\pi} | r(\tilde{\pi}) = r(\pi)] \right)^{1-\lambda} \gamma_\pi(d\pi). \quad (6)$$

Notice that here the measure with respect to which the integral is taken is simply $\gamma_\pi(d\pi)$, which is *independent* of r . This means that, fixing λ , the loss in information can be thought of as coming exactly from the pre-evaluation of the function $\pi^\lambda (1 - \pi)^{1-\lambda}$. Since, for any λ , this function is concave, information loss is exactly due to

¹⁰Although γ_π is the measure of posteriors on state L , I use γ_π to indicate that this is the ex-ante measure, and not the measure conditional on state L .

Jensen's inequality. The interpretation is exactly the same as the intuition as before: it is more efficient to separate an informative signal and an uninformative signal than it is to have them grouped into a moderately informative signal. Proposition 4 gives a way to exactly quantify this trade-off.

This analysis can be used to explain why increases in homogeneity increase the asymmetry of the optimal review system. In particular, the larger the spread of the posteriors within a bin, the more information will be lost within that bin, because the larger will be the difference in Jensen's inequality. For small values of μ , different values of α lead to very different distributions of posteriors $d\gamma_\pi$. As observed in Figure 4a, as α increases, simultaneously more signals are very uninformative and more signals are very informative. This is confirmed in Figure 4b, where the distributions of posteriors are plotted for the same values of α . As α increases, the spread in the distribution of posteriors increases, driving the increase in $\kappa_\alpha(0)$. At the same time, the larger mass near certainty (mass near 0 and 1) incentivizes moving the threshold away from symmetry.

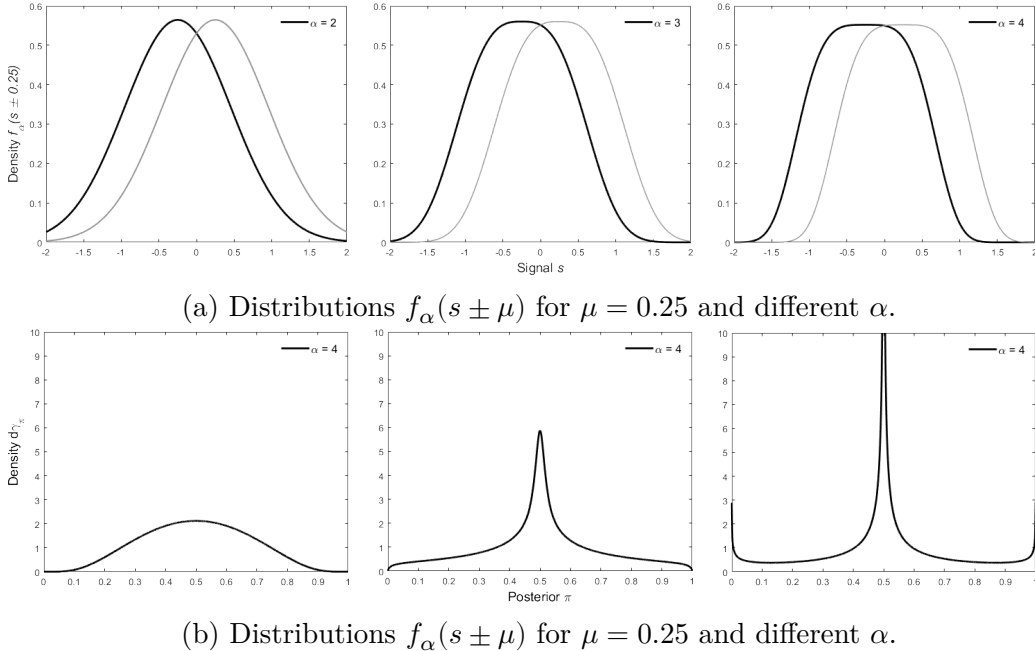


Figure 4: Different distributions of $f_\alpha(s \pm \mu)$ and corresponding distributions of posteriors for $\mu = 0.25$ and different values of α .

6 Extensions

6.1 Non-Uniform Reporting Rates

In the previous sections I assume that given a review system r , the rate of reporting p_r is uniform across the population of reviewers. This restriction is made for tractability; in practice a reviewer's experience is likely to affect their likelihood of submitting a review. It is reasonable that a reviewer with an extreme review is more likely to leave a review, and in some contexts reviews may differentially report positive or negative experiences.¹¹ In this section I show how to incorporate non-uniform reporting rates into the framework studied above. While the qualitative insights from previous sections remain, several new forces emerge in this setting.

Suppose now that a reviewer with signal s facing review r has a probability of leaving a review given by $p(r, s)$, dependent on both the review system *and* the reviewer's signal. As before, it is the difference in measures for each review for the two states that is important for the rate of learning of the platform. In this setting, however, it is not possible to separate the propensity of reviewing from the underlying signal distribution. Explicitly, define the measure $\gamma_L^{(p,r)}$ of reviews in state L (implicitly dependent on μ) as

$$\gamma_L^{(p,r)}(B) := \int_{\{s:r(s) \in B\}} p(r, s) f(s + \mu) ds.$$

Similarly define the measure $\gamma_H^{(p,r)}$ of reviews in state H . As before, these measures are used to determine the rate at which the platform learns the state (the analogue of $p_r \nu(r)$) in this setting:

$$\rho(\mu; r, p) := 1 - \min_{\lambda \in [0,1]} \left[\int_{R_r} \left(\frac{d\gamma_L^{(p,r)}}{d\gamma_H^{(p,r)}} \right)^\lambda d\gamma_H^{(p,r)} + \left(\gamma_L^{(1-p,r)}(\mathbb{R}) \right)^\lambda \left(\gamma_H^{(1-p,r)}(\mathbb{R}) \right)^{1-\lambda} \right].$$

There are two significant differences between ρ and $p_r \nu(r)$. First, there is a mechanical change: since $p(r, \cdot)$ varies with the signal realization, different signals

¹¹For instance, Lafky (2014) suggests that individuals are more likely to report extreme experiences than moderate experiences because they derive a benefit from informing others, and extreme experiences are more informative.

receive different weight, which is why γ_θ now depends non-trivially on p . Second, there is a conceptual change: “null reviews” (reviews that are not submitted) can now be informative of the state. This is the contribution of the second term. Null reviews are effectively an additional signal: if reviewers are more likely to be submitted when the state is L , a lack of reviews is an indication that the state is H .

Theorem 1 generalizes immediately to this setting, where now the ρ ’s take the role of the $p_r\nu(r)$.

Lemma 3. *Fix two review systems r, r' with such that $\rho(\mu; r, p) > \rho(\mu; r', p)$. Fixing a decision problem for the platform, there exists an $\bar{N}$ such that for all $N \geq \bar{N}$, such that $u^*(r, N) > u^*(r', N)$.*

While Lemma 3 shows that even with full flexibility in reporting rates a generic ordering over review systems still exists, without additional structure on the propensities to review p , little more can be said about the comparison. When arbitrary functions p are allowed, then it is as if the underlying distribution of signals that the review systems are drawing from differ. If these distributions differ, then structure is lost and anything can happen. In order to overcome this while still gaining some additional flexibility over the assumption of uniform reporting rates, I restrict attention to functions p that are separable between the impact of the review function and the signal realization.

Definition 3. The propensity to review function $p(r, s)$ is said to be *separable* if $p(r, s) = p_r q(s)$ for some function q .

This specification represents a two-stage procedure. The first stage reflects individuals who are willing to ever submit a review of type r (of which there is a proportion p_r). In the second stage, individuals who might submit a review of this type make their decision based off of their signal realization: among those with signal s only proportion $q(s)$ will actually leave the review.¹² This reflects that even reviewers who sometimes leave reviews will not always leave them, and that their choice may depend on their signal.¹³

¹²A micro-foundation for this two-stage procedure is given in Appendix A.4.1.

¹³Importantly, q depends on the signal and *not* the taste shock of the individual. This latter case would simply reduce to the setting of the previous sections.

In order to gain an understanding of the new forces that enter when reporting rates are not uniform, I now develop an analogue of Theorem 2 in the case that propensities to review are separable. Before I present the result, it is necessary to present some additional notation. Fix the distribution of taste heterogeneity f and the signal reporting rates q . Define the functional E as the integral of a function g with respect to the measure whose density is given by $q \cdot f$:

$$E(g) := \int_{-\infty}^{\infty} g(s)q(s)f(s)ds.$$

Note that E is not an expectation because in general $q \cdot f$ will not be a probability distribution. Similarly, given a review function r , and a review $\tilde{r} \in R_r$, denote by

$$E(g; \tilde{r}) := \int_{-\infty}^{\infty} g(s)\mathbb{1}\{r(s) \in \tilde{r}\}q(s)f(s)ds$$

the integral with respect to the restricted measure. With this notation, Theorem 2 generalizes to the following result.

Theorem 3. *Suppose that f satisfies Assumptions 1 and 2 and $p(r, s) = p_r q(s)$ where q is continuous. Let $\ell_f := \frac{f'}{f}$ denote the linear score of f . Then learning-loss for a finite review system r is given by*

$$\frac{p_r}{p_f} \cdot \lim_{\mu \rightarrow 0} \frac{\rho(\mu; p_f \cdot q)}{\rho_q(\mu; r, p_r \cdot q)} = \frac{E(\ell_f^2) + \frac{p_f E(\ell_f)^2}{1 - p_f E(1)}}{\sum_{\tilde{r} \in R_r} \frac{E(\ell_f; \tilde{r})^2}{E(1; \tilde{r})} + \frac{p_r E(\ell_f)^2}{1 - p_r E(1)}}. \quad (7)$$

The left-hand side of (7) is just the generalization of κ from Section 4.2. In the case that $q \equiv 1$ and r is a threshold rule, Theorem 3 is a generalization of Theorem 2 to multiple bins. The mechanical change discussed above is the reason for the increased notational burden, and reflected in all of the integrals being taken with respect to the new measure with density $q \cdot f$ instead of simply f . The first terms of the numerator and denominator are generalizations from Theorem 2.

The impact of null reviews is reflected in the second term of the numerator and denominator. Importantly, the information contained in null reviews is related to the rate at which a review is submitted. In this setting, the review system's performance

is *dependent* on how frequently it is reported. To see this, consider Figure 5. The top two panels show the distribution of submitted reviews. As p_r increases, these distributions are simply scaled upward, and so separability holds here. However, this is not the case for null reviews, which are shown in the bottom panel. As the review system causes fewer reviewers to report ($p_r \rightarrow 0$), the mass of null reviews becomes similar in the two states, since $1 - q(s)p_r$ approaches 1 uniformly (drowning out any asymmetry due to q). This means that null reviews are less informative as p decreases. In particular, in cases when few people submit reviews (i.e., when p_r is small), the intuition from uniform reporting rates applies almost exactly to the setting with non-uniform reporting rates since null reviews are not informative.

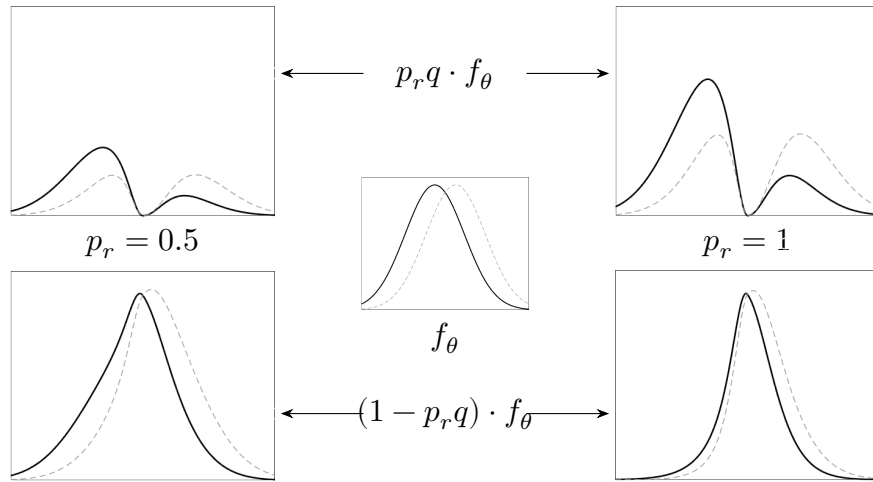


Figure 5: Distributions of reviews for $f_\theta(s)$ normal, with $\mu = 0.25$, and $q(s) = 1 - \frac{1}{1+2(x^2+4\mathbb{1}_{\{x \leq 0\}}x^2)}$ and for $p_r = 0.5$ (left), 1 (right). The top panels show the distribution of submitted reviews, and the bottom panels show the distribution of null reviews.

6.2 General Finite Review Systems

As Theorem 3 shows, it is possible to generalize the results of Theorem 1 to the case when there are more than two bins.

Corollary 2. *Fix an f that satisfies Assumptions 1 and 2, suppose that reporting rates are uniform, and let r_b be a review system that is defined by thresholds $-\infty =$*

$\tau_0 \leq \tau_1 \leq \dots \leq \tau_b = \infty$. Then,

$$\lim_{\mu \rightarrow 0} \frac{\nu(\mu)}{\nu(\mu; r_b)} = I_f / \left(\sum_{i=1}^b \frac{(f(\tau_i) - f(\tau_{i-1}))^2}{F(\tau_i) - F(\tau_{i-1})} \right).$$

In order to solve for the optimal threshold for a review system with b bins, $b - 1$ equations must be solved. Given a distribution f , it is straightforward to do this numerically. See Figure 6 for the optimal three-bin system for different degrees of homogeneity.

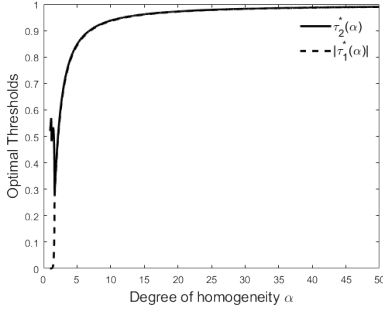
Similar qualitative results appear here as in the binary system. In particular, information loss is increasing in the degree of homogeneity, and the optimal asymmetry of the system is also increasing in the degree of homogeneity. An important difference here over the binary system is that *both* sides of the distribution can be isolated. What is important is that as homogeneity grows, it is increasingly important to effectively “throw-out” large portions of the population. The third bin does not provide any information to the platform (since it is equally likely in each state). Its presence is important, however, because it allows the platform to effectively skim uninformative reviews in order to isolate strong signals on both sides of the distribution.

6.3 Relationship between Production and Consumption of Information

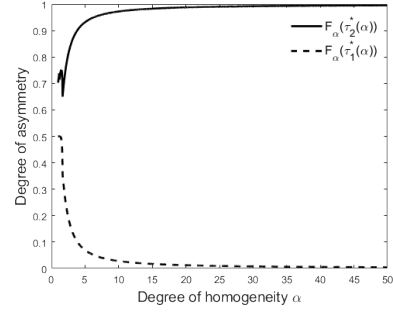
As discussed above, the rates of reporting p (whether uniform or not) are taken from an ex-ante perspective, and not an ex-post perspective. An ex-post perspective would be to directly compare n_r reviews of type r and $n_{r'}$ reviews of type r' .¹⁴ In practice, this latter decision falls to consumers who must decide between multiple types of reviews on a website. While the ex-ante and ex-post preferences over sources of information are related, they are not identical.

Consider two different review systems, r and r' , with $\nu(r) \gg \nu(r')$. If $p_r \ll p_{r'}$, then the platform may be worried about receiving any reviews of type r , and so might prefer r' . However, on average, there will still be many reviews of type r submitted, and so to a consumer deciding between Np_r reviews from system r and $Np_{r'}$ reviews

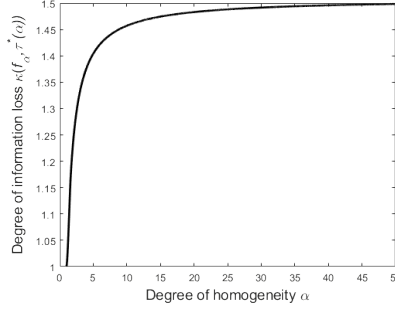
¹⁴This is the perspective that has traditionally been taken in the literature. For instance, see Chernoff (1952), Moscarini and Smith (2002), and Mu et al. (2021).



(a) Optimal thresholds as a function of homogeneity.



(b) Optimal degree of asymmetry, as a function of α .



(c) Minimal information loss, as a function of homogeneity.

Figure 6: Plots for optimally chosen three-bin review system.

from system r' , the “better” reviews might seem better. This intuition is correct: the ex-post perspective leads to a stronger preference for finer reviews. However, in the context of online reviews, when signals are uninformative (so that $\nu(r) \approx 0$), this difference disappears and the two orders agree. Appendix B formalizes the model and the differences, and shows these relationships.

7 Conclusion

In this paper I quantify, in a rate of learning sense, how much information is lost when signals are coarsened. I provide several characterizations of this quantity, both in one-dimensional signal and posterior space. I apply the framework to the context of online reviews and find that the degree of taste homogeneity that the product being reviewed induces in the population has important implications for how much information is lost when signals are coarsened and for the structure of optimal binary systems. As the population becomes more homogeneous, more information is lost. A symmetric binary

system can lose an arbitrary amount of information relative to the full signal, but this loss can be greatly mitigated by optimally using an asymmetric system. The same qualitative results extend to other review systems with finite bins. These results explain why certain review systems are used in practice—in particular explaining the prevalence of positively-skewed systems and binary schemes—and highlight what forces platforms should consider when designing review systems.

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A Proofs

All proofs are contained in this appendix, broken out by section.

A.1 Proofs for Section 3

In this appendix I provide the proof of the major result from Section 3. Before I prove the main result, I first state the following result from Torgersen (1981), which provides the foundation for my result:

Theorem (Torgersen 1981). *Suppose that reviews are always submitted (i.e., $p_r = 1$ for all review systems r). Fix two review systems r, r' with $\nu(r) > \nu(r')$ and a decision problem for the platform. There exists an $\bar{N}$ such that for all $N \geq \bar{N}$, such that $u^*(r, N) > u^*(r', N)$.*

Proof of Theorem 1. For ease of notation, write $\nu_r := \nu(r)$ to denote the learning efficiency of a review r . The pair (p_r, r) can be thought of as defining a statistical experiment over the two states of the world.

In order to apply the above theorem to this setting, we must translate (p_r, r) into a statistical experiment without stochastic reporting, and show that its learning efficiency is given by $p_r \nu_r$, where ν_r is the learning efficiency of the statistical experiment $(1, r)$.

Let $\gamma_L^{(p_r, r)}$ be the measure defined over $R_r \cup \{\emptyset\}$ representing the distribution of reviews in state L . That is,

$$\begin{aligned}\gamma_L^{(p_r, r)}(B) &:= p_r \cdot \int_{\{s: r(s) \in B\}} f_L(s) ds + (1 - p_r) \cdot \mathbb{1}\{\emptyset \in B\} \\ &= p_r \cdot \gamma_L^r(B \cap R_r) + (1 - p_r) \cdot \mathbb{1}\{\emptyset \in B\}.\end{aligned}$$

The only difference between this measure and the measure γ_L^r is the weight $1 - p_r$ placed on $\{\emptyset\}$, which represents a null review (a reviewer choosing to not submit a review). Similarly,

$$\gamma_H^{(p_r, r)}(B) := p_r \cdot \gamma_H^r(B \cap R_r) + (1 - p_r) \cdot \mathbb{1}\{\emptyset \in B\}.$$

Notice then, for a given value of λ , it is the case that

$$\begin{aligned}\int_{R_r \cup \{\emptyset\}} \left(d\gamma_L^{(p_r, r)}\right)^\lambda \left(d\gamma_H^{(p_r, r)}\right)^{1-\lambda} &= \int_{R_r} (p_r \cdot d\gamma_L^r)^\lambda (p_r \cdot d\gamma_H^r)^{1-\lambda} + (1 - p_r) \\ &= p_r \int_{R_r} (d\gamma_L^r)^\lambda (d\gamma_H^r)^{1-\lambda} + 1 - p_r.\end{aligned}$$

Hence,

$$\min_{\lambda \in [0, 1]} \int_{R_r \cup \{\emptyset\}} \left(d\gamma_L^{(p_r, r)}\right)^\lambda \left(d\gamma_H^{(p_r, r)}\right)^{1-\lambda} = 1 - p_r \left(1 - \min_{\lambda \in [0, 1]} \int_{R_r} (d\gamma_L^r)^\lambda (d\gamma_H^r)^{1-\lambda}\right).$$

Concluding,

$$\nu_{(p_r, r)} = 1 - \min_{\lambda \in [0, 1]} \int_{R_r \cup \{\emptyset\}} \left(d\gamma_L^{(p_r, r)}\right)^\lambda \left(d\gamma_H^{(p_r, r)}\right)^{1-\lambda} = p_r \cdot \nu_r,$$

which is what we set out to show. $\square$

A.2 Proofs for Section 4

In this Appendix I provide the proofs of the results in Section 4.

Proof of Lemma 1. This follows from the Monotone Likelihood Ratio Property of log-concave densities and Proposition 5, which is stated and proved in Appendix A.3. $\square$

Proof of Theorem 2. This follows from Theorem 3 in the case that r is a threshold rule and $q \equiv 1$. $\square$

To begin the analysis, some preliminary calculations are needed. Closely related to our family interest is the Gamma function.

Definition 4. Denote by $\Gamma(\alpha)$ the standard Gamma function, given by

$$\Gamma(\alpha) := \int_0^\infty x^{\alpha-1} e^{-x} dx.$$

Let $\Gamma(\alpha, t)$ denote the upper incomplete gamma function:

$$\Gamma(\alpha, t) := \int_t^\infty x^{\alpha-1} e^{-x} dx.$$

Finally, let $\gamma(\alpha, t)$ denote the lower incomplete gamma function

$$\gamma(\alpha, t) := \int_0^t x^{\alpha-1} e^{-x} dx = \Gamma(\alpha) - \Gamma(\alpha, t).$$

Lemma 4. *It is the case that*

$$f_\alpha(x) = \frac{\alpha}{2\Gamma(1/\alpha)} e^{-|x|^\alpha}.$$

Moreover, the Fisher information of f_α is given by

$$I_{f_\alpha}(0) = \frac{\alpha^2}{\Gamma(1/\alpha)} \Gamma\left(2 - \frac{1}{\alpha}\right).$$

Proof of Lemma 4. The first part of the statement is just ensuring that f_α is indeed a probability distribution.

$$\int_{-\infty}^\infty e^{-|x|^\alpha} dx = 2 \int_0^\infty e^{-x^\alpha} dx.$$

Make now the substitution $u = x^\alpha$. Then, $\frac{1}{\alpha} u^{\frac{1}{\alpha}-1} du = dx$ so that

$$2 \int_0^\infty e^{-x^\alpha} dx = 2 \int_0^\infty \frac{1}{\alpha} u^{\frac{1}{\alpha}-1} e^{-u} du = \frac{2}{\alpha} \Gamma\left(\frac{1}{\alpha}\right).$$

This proves the first claim. To show the second claim, recall that the Fisher Information is also equal to

$$I_f = \mathbb{E} \left[\left(\frac{f'(s)}{f(s)} \right)^2 \right] = -\mathbb{E} \left[\frac{\partial^2}{\partial s^2} \log(f(s)) \right].$$

This means that it is the case that

$$\begin{aligned} I_{f_\alpha} &= \frac{\alpha}{2\Gamma(1/\alpha)} \int_{-\infty}^{\infty} \left(\frac{\partial^2}{\partial x^2} |x|^\alpha \right) e^{-|x|^\alpha} dx = \frac{\alpha}{\Gamma(1/\alpha)} \int_0^{\infty} \left(\frac{\partial^2}{\partial x^2} x^\alpha \right) e^{-x^\alpha} dx \\ &= \frac{\alpha}{\Gamma(1/\alpha)} \int_0^{\infty} \alpha(\alpha-1)x^{\alpha-2} e^{-x^\alpha} dx. \end{aligned}$$

Now, perform again the substitution of $u = x^\alpha$.

$$\begin{aligned} &= \frac{\alpha}{\Gamma(1/\alpha)} \int_0^{\infty} \alpha(\alpha-1) \frac{1}{\alpha} u \cdot u^{-\frac{2}{\alpha}} \cdot u^{\frac{1}{\alpha}-1} e^{-u} dx \\ &= \frac{\alpha}{\Gamma(1/\alpha)} \int_0^{\infty} (\alpha-1) u^{-\frac{1}{\alpha}} e^{-u} dx = \frac{\alpha(\alpha-1)}{\Gamma(1/\alpha)} \Gamma\left(1 - \frac{1}{\alpha}\right). \end{aligned}$$

Since it is the case that $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$,

$$\Gamma\left(1 - \frac{1}{\alpha}\right) = \frac{\alpha}{\alpha-1} \Gamma\left(2 - \frac{1}{\alpha}\right),$$

Concluding,

$$I_{f_\alpha}(0) = \frac{\alpha^2}{\Gamma(1/\alpha)} \Gamma\left(2 - \frac{1}{\alpha}\right).$$

□

In all proofs, denote $\kappa_\alpha(\tau)$ by $\kappa(\alpha; \tau)$. This is just done for clarity: κ is a function of α .

Proof of Lemma 2. First, a simple computation shows that $\kappa(\alpha; 0) = \Gamma\left(\frac{1}{\alpha}\right) \Gamma\left(2 - \frac{1}{\alpha}\right)$. This shows that $\kappa(1; 0) = 1$, as $\Gamma(1) = 1$.

I now proceed to the other claims. Let $\psi(\alpha)$ denote the digamma function

$$\psi(\alpha) := \frac{\Gamma'(\alpha)}{\Gamma(\alpha)}.$$

It is well-known that ψ is strictly increasing on $(0, \infty)$.

Now, let $\kappa'(\alpha; 0)$ denote $\frac{\partial}{\partial \alpha} \kappa(\alpha; 0)$. A straightforward computation yields that

$$\kappa'(\alpha; 0) = \frac{\Gamma\left(\frac{1}{\alpha}\right) \Gamma\left(2 - \frac{1}{\alpha}\right)}{\alpha^2} \left(\psi\left(2 - \frac{1}{\alpha}\right) - \psi\left(\frac{1}{\alpha}\right) \right).$$

Since ψ is strictly increasing, it is immediately seen that $\kappa'(\alpha; 0) > 0$.

It is left to show that $\lim_{\alpha \rightarrow \infty} \kappa(\alpha; 0) = \infty$. First, it can be easily verified that $\psi(2) < 1$ and $\lim_{\alpha \rightarrow \infty} \frac{1}{\alpha} \psi\left(\frac{1}{\alpha}\right) = -1$. These two facts together show that κ' can be seen as

$$\frac{\kappa'(\alpha; 0)}{\kappa(\alpha; 0)} \approx \frac{1}{\alpha}.$$

That is, $\lim_{\alpha \rightarrow \infty} \alpha \frac{\kappa'(\alpha; 0)}{\kappa(\alpha; 0)} = 1$. This means that $\frac{\kappa(\alpha; 0)}{\alpha} \rightarrow 1$, and hence $\kappa'(\alpha; 0) \rightarrow 1$. This shows that κ diverges, completing the proof. $\square$

Proof of Proposition 1. As noted, I restrict attention to $\tau \in [0, \infty)$ for the proof. The exact analogues apply for $\tau \leq 0$.

For ease of reference I restate that the objective that the platform aims to maximize is given by.

$$g_\alpha(\tau) := \frac{f_\alpha(\tau)^2}{F_\alpha(\tau)(1 - F_\alpha(\tau))} \tag{8}$$

I proceed in several steps. I first show that $\tau^*(\alpha) \neq 0$ for $\alpha > 2$, and that 0 is a candidate for $\tau^*(\alpha)$ for $\alpha \leq 2$.

Lemma 5. *It is the case that 0 is a local maximum of $g_\alpha(\tau)$ for $\alpha \leq 2$, and a local minimum of $g_\alpha(\tau)$ for $\alpha > 2$.*

Proof of Lemma 5. The derivative of (8) with respect to τ is given by (dropping the

subscript α)

$$\frac{f(\tau)}{F(\tau)(1-F(\tau))} \left[2f'(\tau) - f(\tau)^2 \left(\frac{1}{F(\tau)} - \frac{1}{1-F(\tau)} \right) \right].$$

As f is of full support, $\frac{f(\tau)}{F(\tau)(1-F(\tau))} > 0$, and so can be ignored when searching for a critical value, and hence a maximum. The first order condition to be a local maximum then becomes

$$2f'(\tau) = f(\tau)^2 \left(\frac{1}{F(\tau)} - \frac{1}{1-F(\tau)} \right). \quad (9)$$

For any $\alpha > 1$, 0 will be a solution to (9), and hence a candidate for a maximizer. This is as $f'_\alpha(0) = 0$ and $F_\alpha(0) = \frac{1}{2}$.

This means that to verify the claim the second order condition is needed. At 0, it can be verified that the second-order condition for a maximizer (since $F_\alpha(0) = \frac{1}{2}$) is

$$\frac{1}{2}f_\alpha(0)^4 + \frac{1}{8}f'_\alpha(0)^2 + \frac{1}{8}f_\alpha(0)f''_\alpha(0) \leq 0.$$

Notice, for all $\alpha > 2$, we have that $f'_\alpha(0) = 0$ and $f''_\alpha(0) = 0$. This means that, as $f_\alpha(0) > 0$, for all $\alpha > 2$, $\tau = 0$ is a local minimum, and hence is not optimal. Similarly, for $\alpha < 2$, it is the case that $\lim_{\tau \rightarrow 0} f''_\alpha(\tau) = -\infty$, while $f'_\alpha(0), f_\alpha(0) > 0$. This means that 0 is indeed a local maximum for $\alpha < 2$. The case $\alpha = 2$ can be computed numerically, and it can be seen that this is indeed a local maximum in this case. $\square$

I now show that there is at most one possible candidate for the maximizer of $g_\alpha(\tau)$ on $\tau \in [0, \infty)$. This will show uniqueness of $\tau^*(\alpha)$.

Lemma 6. *The following are true:*

- (i) *For $\alpha > 2$, there is exactly one value of $\tau \in (0, \infty)$ that satisfies (9) and it is a local maximum of $g_\alpha(\tau)$; and*
- (ii) *For $\alpha \leq 2$, no value of $\tau \in (0, \infty)$ satisfies (9).*

Proof of Lemma 6. Throughout I drop the subscript of α for notational clarity.

Case 1: $\alpha > 2$.

Straightforward algebra allows (9) to be rewritten as

$$\frac{2f'(\tau)}{1 - 2F(\tau)} = \frac{f^2(\tau)}{F(\tau)(1 - F(\tau))}. \quad (10)$$

Notice that the right-hand side of (10) is exactly $g(\tau)$. This means that the value of g can also be used to determine the sign of its derivative. Specifically, the critical points of the objective are exactly those that intersect the curve given by

$$w(\tau) := \frac{2f'(\tau)}{1 - 2F(\tau)}.$$

It is exactly when $g(\tau) = w(\tau)$ that $g'(\tau) = 0$. Moreover,

$$g'(\tau) > 0 \iff g(\tau) > w(\tau).$$

Due to this relationship, the characteristics of $w(\tau)$ imply certain characteristics of $g(\tau)$. In particular, I now show that $w(\tau)$ has one unique maximum on $(0, \infty)$. To do this, I iterate on the process above. The sign of $w'(\tau) = 0$ is determined by the sign of

$$2f''(\tau)(1 - 2F(\tau)) + 4f'(\tau)f(\tau) \quad (11)$$

in particular, $w'(\tau) = 0$ if and only if

$$\frac{2f'(\tau)}{1 - 2F(\tau)} = -\frac{f''(\tau)}{f(\tau)} =: h(\tau).$$

Moreover, $w'(\tau) > 0$ if and only if $h(\tau) > w(\tau)$ (notice that this is the reverse relationship of w and g).

While w and g are difficult to manage because the presence of F in their definition (and there is no closed-form expression for the incomplete Gamma functions), $h(\tau) = (\alpha^2 - \alpha)x^{\alpha-2} - \alpha^2x^{2\alpha-2}$. This (relative to g and w) is an extremely simple function. It can be seen that for all $\alpha > 2$, h has the following properties: (i) $h(0) = 0$, and $h(\tau) > 0$ for τ in a neighbourhood of 0; (ii) h is single-peaked; (iii) $h(1) < 0$; and

(iv) $h(\tau) > w(\tau)$ in a neighbourhood of 0. Properties (i)-(iii) are easily to verify, and property (iv) follows from $w'(\tau) > 0$ in a neighbourhood of 0 (which directly implies that $w(\tau) < h(\tau)$).

Claim 1a: $w(\tau)$ has at one critical value on $(0, \infty)$. It is a local maximum, and this critical value is on $(0, 1)$.

Recall that if $w(\tau) < h(\tau)$, then $w'(\tau) > 0$, and if $w(\tau) > h(\tau)$, then $w'(\tau) < 0$. First, observe that all critical values of w must be on $(0, 1)$ because h is negative of $[1, \infty)$ (so $w(\tau) \neq h(\tau)$ for any $\tau \geq 1$).

Second, let τ_h be the value of τ that maximizes h . Notice that h is increasing on $(0, \tau_h)$ and decreasing on (τ_h, ∞) . Suppose now that $w(\tau') = h(\tau')$ for some $\tau' \in (0, \tau_h)$. There must be smallest value of τ' for which this holds, as $w(\tau)$ is strictly increasing in a neighbourhood of 0. Then for this τ' , it is the case that $w(\tau') = 0$ and $h(\tau') > 0$. This means that for some $\epsilon > 0$ it is the case that for all $\tau \in (\tau' - \epsilon, \tau')$ it is the case that $w(\tau) > h(\tau)$. But τ' is the smallest positive value where the two cross, which is a contradiction. This means that it cannot be the case that $w'(\tau) = 0$ when $h'(\tau) > 0$. This is as, for w to cross h when h is increasing it must be the case that w is larger than h before the crossing.

Since eventually $w(\tau) > h(\tau)$, it must be the case then that either $w = h$ at the peak of h , or on the region where h is decreasing. In either case it must be the case that w goes from increasing to decreasing, and can not cross h again. This proves the claim.

Claim 1b: $g(\tau)$ has a most one critical value on $(0, \infty)$. It is a local maximum, and this critical value is on $(0, 1)$.

Recall that, if $g(\tau) > w(\tau)$, then $g'(\tau) > 0$, and if $g(\tau) < w(\tau)$, then $g'(\tau) < 0$. Notice that because zero is a local minimum of $g(\tau)$ (by Lemma 5), $g(\tau)$ is increasing in a neighbourhood of zero (since we are concerned only with $\tau \geq 0$). Moreover, it is the case that $g(0) > 0 = w(0)$. This second equality holds because $f''(0) = 0$ for $\alpha > 2$. Denote by τ_w the value of τ that maximizes w on $(0, \infty)$.

Now, suppose that $g(\tau') = w(\tau')$ for some $\tau' \in (\tau_w, \infty)$. This is because it implies that $g(\tau) > w(\tau)$ for $\tau = (\tau', \tau' + \epsilon)$ for some $\epsilon > 0$. However, this means that $g'(\tau) > 0$ on this region. Because w is decreasing on this region, it implies that $g'(\tau) > 0$ on (τ', ∞) , contradicting that $\lim_{\tau \rightarrow \infty} g(\tau) = 0$. This means that the two

cannot intersect when $w'(\tau) < 0$.

By a similar logic, it must be the case that $g(\tau) = w(\tau)$ for some $\tau \in (0, \tau_w]$. Call the first point where they intersect τ^* (well-defined because they are not equal at 0). Notice that for all $\tau \in (\tau^*, \tau^* + \epsilon)$, it must be the case that g is decreasing.¹⁵ But, g must continue to decrease until they intersect again. So, they cannot intersect until w decreases. We have, however, ruled out intersection in that case, so it must be that τ^* is unique, proving the claim and finishing Case 1.

Case 2: $\alpha < 2$.

In the case of $\alpha < 2$ showing that there is no critical value on $(0, \infty)$ is simpler and so I omit the details.

In this case, (i) $\lim_{\tau \rightarrow 0} w(\tau) = \infty$ and (ii) $h(\tau)$ is monotonically decreasing. These jointly imply that $w(\tau)$ is monotonically decreasing to 0. This in turn implies that $g(\tau)$ is monotonically decreasing to zero (since they cannot intersect when w decreases in a similar argument to above). This shows that in the case that $\alpha < 2$ it must be that $\tau = 0$ is the only possible critical value.

Case 3: $\alpha = 2$.

This is dealt with in a similar manner to Case 2. The major difference is that $w(\tau)$ does not diverge at 0. Instead, it is easy to show that $w(0) = h(0)$ and w is decreasing in a region of 0, so that it must be monotonically decreasing. This case then reduces to Case 2. $\square$

Now that I have showed that τ^* exists and is unique, it is sufficient to look at the solution to the first order condition (9) on $(0, \infty)$. The last things to check are that $\tau^*(\alpha) \rightarrow 1$, and the bounding value of κ .

Claim: $\lim_{\alpha \rightarrow \infty} \tau^*(\alpha) = 1$.

First, notice that $\tau^*(\alpha) < 1$ for all $\alpha \geq 1$, by Lemma 6.

So, I simply show that for all $\tau < 1$, there exists an α_τ such that: $\alpha \geq \alpha_\tau$, $g'_\alpha(\tau) > 0$. To do this, explicitly write the inner term

$$2f'(\tau) - f(\tau)^2 \left(\frac{1}{F(\tau)} - \frac{1}{1 - F(\tau)} \right)$$

¹⁵This is obvious in the case that $\tau^* < \tau_w$. In the case that $\tau^* = \tau_w$, it follows from a similar argument to showing that there is no intersection when w is decreasing.

$$\begin{aligned}
&= -\alpha\tau^{\alpha-1} \cdot \frac{\alpha}{\Gamma(1/\alpha)} e^{-\tau^\alpha} \\
&\quad + \left(\frac{\alpha}{2\Gamma(1/\alpha)} e^{-\tau^\alpha} \right)^2 \left[\frac{\Gamma(1/\alpha)}{\frac{1}{2}\Gamma(1/\alpha, \tau^\alpha)} - \frac{\Gamma(1/\alpha)}{\frac{1}{2}\Gamma(1/\alpha) + \frac{1}{2}\gamma(1/\alpha, \tau^\alpha)} \right],
\end{aligned}$$

where here $\Gamma(1/\alpha, \cdot)$ and $\gamma(1/\alpha, \cdot)$ are the upper- and lower-incomplete gamma functions, respectively. Because we are only interested in the sign of this expression, it is possible to simplify. It is equivalent to look at the sign of

$$-2\tau^{\alpha-1} + e^{-\tau^\alpha} \left[\frac{1}{\Gamma(1/\alpha, \tau^\alpha)} - \frac{1}{\Gamma(1/\alpha) + \gamma(1/\alpha, \tau^\alpha)} \right]. \quad (12)$$

That is, $g'_\alpha(\tau) > 0$ exactly when (12) is positive. When $\tau < 1$, the first term converges to 0, so we need only ensure that the second term stays bounded from 0. First, for fixed $\tau < 1$, $e^{-\tau^\alpha} \rightarrow 1$. Second, as $\alpha \rightarrow \infty$, it is easy to verify that $f_\alpha(\tau) \rightarrow \frac{1}{2}\mathbb{1}\{\tau \in [-1, 1]\}$ (except for the measure zero set of $\{-1, 1\}$). This means that

$$\lim_{\alpha \rightarrow \infty} \left(\frac{1}{F(\tau)} - \frac{1}{1 - F(\tau)} \right) = \left(\frac{2}{\tau + 1} - \frac{2}{1 - \tau} \right) > 0$$

for all fixed $\tau \in (0, 1)$. This means that neither portions of the second term go to zero, so that for any τ in this interval, $g'_\alpha(\tau)$ is eventually positive.

This means that $\tau^*(\alpha)$ must eventually grow to 1. It is not hard to use the same arguments to show that $\tau^*(\alpha)^\alpha$ converges to the solution of $\Gamma(0, x) - \frac{1}{2x}e^{-x} = 0$.

Claim: $\kappa(\alpha; \tau^*(\alpha)) < 2e^2 \int_1^\infty \frac{e^{-x}}{x} dx$.

For all $\alpha \geq 4$, it can be seen that $\kappa(\alpha, 1)$ is less than this limit. For $\alpha < 4$, using the naive threshold of 0 will also return a value that is less than this limit. This shows the final claim, and completes the proof of Proposition 1. $\square$

Proof of Proposition 2. In order to prove Proposition 2(i), I prove the following, stronger, lemma.

Lemma 7. $\tau^*(\alpha)^\alpha$ is strictly increasing in α for $\alpha \geq 2$.

Proof of Lemma 7. First, notice that

$$\frac{\partial \tau^*(\alpha)^\alpha}{\partial \alpha} > 0 \iff \left(\frac{\partial^2}{\partial \tau^\alpha \partial \alpha} \frac{f_\alpha^2(\tau)}{F_\alpha(\tau)(1 - F_\alpha(\tau))} \right) \Big|_{\tau=\tau^*(\alpha)} > 0.$$

That is, letting $x = \tau^\alpha$, the sign of

$$\frac{\partial}{\partial \alpha} \left[-2 \frac{x}{x^{1/\alpha}} e^x + \left[\frac{1}{\Gamma(1/\alpha, x)} - \frac{1}{\Gamma(1/\alpha) + \gamma(1/\alpha, x)} \right] \right] \quad (13)$$

must be determined at the optimal value of $x^*(\alpha) = \tau^*(\alpha)^\alpha$. Notice that (13) is not exactly equal to the derivative, because there is an additional positive term that multiplies this. However, because what is in the brackets here is zero at the optimal threshold, this term can be ignored (because it will not influence sign of this object). This just follows from the envelope theorem. So, we must show that (13) is positive at $x^*(\alpha)$.

Another simplification is to replace α with $1/\alpha$. That is, let $\beta := 1/\alpha$. A straightforward computation that yields that the sign of (13) is the opposite of the sign of

$$b_\beta(x) := 2 \log(x) \frac{x}{x^\beta} e^x - \frac{\int_x^\infty y^{\beta-1} \log(y) e^{-y} dy}{\Gamma(\beta, x)^2} + \frac{\int_0^\infty y^{\beta-1} \log(y) e^{-y} dy + \int_0^x y^{\beta-1} \log(y) e^{-y} dy}{(\Gamma(\beta) + \gamma(\beta, x))^2}.$$

This is because $\frac{\partial}{\partial \alpha} \beta < 0$. So, I show that $b_\beta(x^*(\beta)) < 0$ (abusing notation here writing $x^*(\beta)$) for $\beta < \frac{1}{2}$. As a final choice to make notation easier, let λ_β denote the measure that is induced by $y^{\beta-1} e^{-y}$ (that is, $d\lambda_\beta(y) = y^{\beta-1} e^{-y} dy$). Then, $b_\beta(x)$ can be written as

$$2 \log(x) \frac{x}{x^\beta} e^x - \frac{\int_x^\infty \log(y) d\lambda_\beta(y)}{\Gamma(\beta, x)^2} + \frac{\int_0^\infty \log(y) d\lambda_\beta(y) + \int_0^x \log(y) d\lambda_\beta(y)}{(\Gamma(\beta) + \gamma(\beta, x))^2}. \quad (14)$$

Note that in the notation above, we have that

$$\Gamma(\beta) = \int_0^\infty d\lambda_\beta(y), \quad \Gamma(\beta, x) = \int_x^\infty d\lambda_\beta(y), \quad \text{and} \quad \gamma(\beta) = \int_0^x d\lambda_\beta(y).$$

Now, at $x^*(\beta)$, from (13), it is the case that

$$2x^{1-\beta} e^x = \frac{1}{\Gamma(\beta, x)} - \frac{1}{\Gamma(\beta) + \gamma(\beta, x)}.$$

This means that it is possible to write (14) at $x^*(\beta)$ (I write x instead of $x^*(\beta)$ for conciseness) as

$$\begin{aligned}
& \log(x) \left(\frac{1}{\Gamma(\beta, x)} - \frac{1}{\Gamma(\beta) + \gamma(\beta, x)} \right) \\
& - \frac{\int_x^\infty \log(y) d\lambda_\beta(y)}{\Gamma(\beta, x)^2} + \frac{\int_0^\infty \log(y) d\lambda_\beta(y) + \int_0^x \log(y) d\lambda_\beta(y)}{(\Gamma(\beta) + \gamma(\beta, x))^2}. \\
& = \frac{\log(x) \int_x^\infty d\lambda_\beta(y)}{\Gamma(\beta, x)^2} - \frac{\log(x) \left(\int_0^\infty d\lambda_\beta(y) + \int_0^x d\lambda_\beta(y) \right)}{\Gamma(\beta) + \gamma(\beta, x)} \\
& - \frac{\int_x^\infty \log(y) d\lambda_\beta(y)}{\Gamma(\beta, x)^2} + \frac{\int_0^\infty \log(y) d\lambda_\beta(y) + \int_0^x \log(y) d\lambda_\beta(y)}{(\Gamma(\beta) + \gamma(\beta, x))^2} \\
& = \frac{\int_x^\infty (\log(x) - \log(y)) d\lambda_\beta(y)}{\Gamma(\beta, x)^2} \\
& + \frac{\int_0^\infty (\log(y) - \log(x)) d\lambda_\beta(y) + \int_0^x (\log(y) - \log(x)) d\lambda_\beta(y)}{(\Gamma(\beta) + \gamma(\beta, x))^2}.
\end{aligned}$$

Since $\log$ is an increasing function, it must be that $\int_x^\infty (\log(x) - \log(y)) d\lambda_\beta(y) < 0$ whenever $\beta < \frac{1}{2}$, so that $x^*(\beta) > 0$. Moreover, it must always be the case that $\Gamma(\beta, x) < \Gamma(\beta) + \gamma(\beta, x)$ whenever $x > 0$. This means that at $x^*(\beta)$ it is the case that

$$\frac{\int_x^\infty (\log(x) - \log(y)) d\lambda_\beta(y)}{\Gamma(\beta, x)^2} < \frac{\int_x^\infty (\log(x) - \log(y)) d\lambda_\beta(y)}{(\Gamma(\beta) + \gamma(\beta, x))^2}.$$

This means that (14) evaluated at $x^*(\beta)$ is less than

$$\begin{aligned}
& \frac{\int_x^\infty (\log(x) - \log(y)) d\lambda_\beta(y)}{(\Gamma(\beta) + \gamma(\beta, x))^2} \\
& + \frac{\int_0^\infty (\log(y) - \log(x)) d\lambda_\beta(y) + \int_0^x (\log(y) - \log(x)) d\lambda_\beta(y)}{(\Gamma(\beta) + \gamma(\beta, x))^2} \\
& = \frac{2 \int_0^x (\log(y) - \log(x)) d\lambda_\beta(y)}{(\Gamma(\beta) + \gamma(\beta, x))^2} < 0.
\end{aligned}$$

This proves that $x^*(\beta)$ is decreasing in β , so that $\tau^*(\alpha)^\alpha$ is increasing in α for all $\beta < \frac{1}{2}$. This in turn shows that $\tau^*(\alpha)^\alpha$ is increasing in α for all $\alpha \geq 2$. $\square$

As noted, Proposition 2(i) follows as an immediate corollary of this lemma, since $\tau^*(\alpha) < 1$ for all α .

Proof of Proposition 2(ii):

In what follows, let $P(\alpha, t) := \gamma(\alpha, t)/\Gamma(\alpha)$ denote the *normalized lower incomplete gamma function*.

First, recall that $\tau^*(\alpha)^\alpha$ strictly increasing in α by Lemma 7.

I show the claim separately on two intervals, based off of the behaviour of $f_\alpha(0)$ as a function of α . In what follows, let α^* be defined as the unique value for which $\psi(1/\alpha) + \alpha = 0$. For $\alpha < \alpha^*$, it is the case that $\Gamma(1/\alpha)/\alpha$ is decreasing in α (so that $f_\alpha(0)$ is increasing), and for $\alpha > \alpha^*$ it is the case the $\Gamma(1/\alpha)/\alpha$ is increasing in α (so that $f_\alpha(0)$ is decreasing).

Case 1: $\alpha < \alpha^*$.

First, on this region, for a fixed τ with $0 < \tau < 1$, it must be the case that $f_\alpha(\tau)$ is increasing in α . This is because clearly τ^α is decreasing in α , and this is the region where $\frac{\alpha}{\Gamma(\alpha)}$ is increasing.

Now, $F_\alpha(0) \equiv \frac{1}{2}$, and for all $0 < \tau < \tau'$, it is the case that $f_\alpha(\tau)$ is increasing in α . This means that, for all $\alpha, \alpha' \in (2, \alpha^*)$ with $\alpha < \alpha'$, it is the case that

$$F_\alpha(\tau^*(\alpha)) < F_{\alpha'}(\tau^*(\alpha))$$

From Lemma 7 it is the case that $\tau^*(\alpha') > \tau^*(\alpha)$, so that

$$F_{\alpha'}(\tau^*(\alpha)) < F_{\alpha'}(\tau^*(\alpha'))$$

This shows that $F_\alpha(\tau^*(\alpha))$ is increasing in α on this range, completing the proof in this case.

Case 2: $\alpha > \alpha^*$.

Since we have that $\alpha > \alpha^*$ and we have showed that $\tau^*(\alpha)$ is increasing in α , it must be the case that $f_\alpha(\tau^*(\alpha))$ is decreasing in α for $\alpha > \alpha^*$, as by definition $\Gamma(1/\alpha)/\alpha$ is increasing for all $\alpha > \alpha^*$.

This means that if it is the case that

$$\frac{f_{\alpha}(\tau^*(\alpha))^2}{F_{\alpha}(\tau^*(\alpha))(1 - F_{\alpha}(\tau^*(\alpha)))} \quad (15)$$

is increasing in α , then the claim will hold in this case. This is because, for (15) to be increasing, if the numerator is decreasing then the denominator must be in turn decreasing. This implies that $F_{\alpha}(\tau^*(\alpha))$ is increasing in α , which is what we set out to show. A stronger condition, which is what we will prove, is that

$$\frac{\left(\frac{\alpha}{\Gamma(1/\alpha)}\right)^2}{(1 + P(1/\alpha, \tau))(1 - P(1/\alpha, \tau))} \quad (16)$$

is increasing in α for fixed τ , where $P(1/\alpha, \tau) := \frac{\Gamma(1/\alpha, \tau)}{\Gamma(1/\alpha)}$ is the normalized incomplete Gamma function. This is a stronger condition because it is saying is that if we look at values of τ for each α that make $e^{-\tau^{\alpha}}$ equal, then these values of τ cause this ratio to increase. In particular, consider evaluating (15) at $\tau^*(\alpha)$ for some α . If (16) is increasing in α , then for all $\alpha' > \alpha$ it must be the case that

$$\frac{f_{\alpha'}(\tau^*(\alpha)^{\alpha/\alpha'})^2}{F_{\alpha'}(\tau^*(\alpha)^{\alpha/\alpha'})(1 - F_{\alpha'}(\tau^*(\alpha)^{\alpha/\alpha'}))} > \frac{f_{\alpha}(\tau^*(\alpha))^2}{F_{\alpha}(\tau^*(\alpha))(1 - F_{\alpha}(\tau^*(\alpha)))}.$$

Since by optimality it is the case that

$$\frac{f_{\alpha'}(\tau^*(\alpha'))^2}{F_{\alpha'}(\tau^*(\alpha'))(1 - F_{\alpha'}(\tau^*(\alpha')))} \geq \frac{f_{\alpha'}(\tau^*(\alpha)^{\alpha/\alpha'})^2}{F_{\alpha'}(\tau^*(\alpha)^{\alpha/\alpha'})(1 - F_{\alpha'}(\tau^*(\alpha)^{\alpha/\alpha'}))},$$

this will prove the claim.

Now, to show that (16) is indeed increasing in α . Perform the switch to $\beta := 1/\alpha$, as the in proof of Lemma 7. Then, the above can be rewritten as

$$\begin{aligned} & \frac{\left(\frac{1}{\beta\Gamma(\beta)}\right)^2}{(1 + P(\beta, x))(1 - P(\beta, x))} \\ &= \frac{1}{2\beta^2\Gamma(\beta)} \left(\frac{1}{\Gamma(\beta) + \gamma(\beta, x)} + \frac{1}{\Gamma(\beta) - \gamma(\beta, x)} \right). \end{aligned}$$

Multiplying by 2 and then taking the derivative with respect to β here yields

$$- \frac{2\beta\Gamma(\beta) + \beta^2 \int_0^\infty \log(y) d\lambda_\beta(y)}{(\beta^2\Gamma(\beta))^2} \left(\frac{1}{\Gamma(\beta) + \gamma(\beta, x)} + \frac{1}{\Gamma(\beta) - \gamma(\beta, x)} \right) \\ - \frac{1}{\beta^2\Gamma(\beta)} \left(\frac{\int_0^\infty \log(y) d\lambda_\beta(y) + \int_0^x \log(y) d\lambda_\beta(y)}{(\Gamma(\beta) + \gamma(\beta, x))^2} + \frac{\int_x^\infty \log(y) d\lambda_\beta(y)}{(\Gamma(\beta) - \gamma(\beta, x))^2} \right).$$

Here I use the definition of $\lambda_\beta(y)$ from the proof of Lemma 7. The goal is to show that this object is negative. Again let $\Gamma'(\beta)/\Gamma(\beta) =: \psi(\beta)$ and observe that for $\alpha > \alpha^*$ by definition it the case that $\frac{1}{\beta} + \psi(\beta) < 0$. Now, straightforward arithmetic shows that the sign of this is equivalent to the sign of

$$- \frac{\int_0^\infty \left(\log(y) + \frac{2}{\beta} + \psi(\beta) \right) d\lambda_\beta(y) + \int_0^x \left(\log(y) + \frac{2}{\beta} + \psi(\beta) \right) d\lambda_\beta(y)}{(\Gamma(\beta) + \gamma(\beta, x))^2} \\ - \frac{\int_x^\infty \left(\log(y) + \frac{2}{\beta} + \psi(\beta) \right) d\lambda_\beta(y)}{(\Gamma(\beta) - \gamma(\beta, x))^2}.$$

I first show that for all $x \in (0, \infty)$ and β , it is the case that.

$$- \int_x^\infty \left(\log(y) + \frac{2}{\beta} + \psi(\beta) \right) d\lambda_\beta(y) < 0.$$

Now, consider that on this region of β , by definition it is the case that

$$- \left(\frac{2}{\beta} + \psi(\beta) \right) = 2 \left(\frac{1}{\beta} + \psi(\beta) \right) + \psi(\beta) < \psi(\beta).$$

This means that

$$- \int_x^\infty \left(\log(y) + \frac{2}{\beta} + \psi(\beta) \right) d\lambda_\beta(y) < \int_x^\infty (-\log(y) + \psi(\beta)) d\lambda_\beta(y)$$

Now, the sign of the above is equivalent to the sign of

$$\frac{\int_x^\infty (-\log(y) + \psi(\beta)) d\lambda_\beta(y)}{\int_x^\infty d\lambda_\beta(y)}.$$

This, however, is clearly decreasing in x , because smaller values of x correspond to

the average log value being smaller. Since it is clearly equal to zero at $x = 0$, this shows that

$$-\int_x^\infty \left(\log(y) + \frac{2}{\beta} + \psi(\beta) \right) d\lambda_\beta(y) < 0.$$

This in turn implies that

$$-\frac{\int_x^\infty \left(\log(y) + \frac{2}{\beta} + \psi(\beta) \right) d\lambda_\beta(y)}{(\Gamma(\beta) - \gamma(\beta, x))^2} < -\frac{\int_x^\infty \left(\log(y) + \frac{2}{\beta} + \psi(\beta) \right) d\lambda_\beta(y)}{(\Gamma(\beta) + \gamma(\beta, x))^2}.$$

This means that it is sufficient to check the sign of

$$\begin{aligned} & -\int_0^\infty \left(\log(y) + \frac{2}{\beta} + \psi(\beta) \right) d\lambda_\beta(y) - \int_0^x \left(\log(y) + \frac{2}{\beta} + \psi(\beta) \right) d\lambda_\beta(y) \\ & - \int_x^\infty \left(\log(y) + \frac{2}{\beta} + \psi(\beta) \right) d\lambda_\beta(y) \end{aligned}$$

and ensure that it is negative. But this expression is equal to

$$-2 \int_0^\infty \left(\log(y) + \frac{2}{\beta} + \psi(\beta) \right) d\lambda_\beta(y) = -\Gamma'(\beta) - \left(\frac{2}{\beta} + \psi(\beta) \right) \Gamma(\beta) < 0,$$

where this last inequality follows from the observation above:

$$\left(\frac{2}{\beta} + \psi(\beta) \right) \Gamma(\beta) > -\Gamma'(\beta)$$

This completes the proof in this case. □

Proof of Proposition 3. First, notice that

$$f(s) := f_{\alpha_L, \alpha_H}(s) = \frac{1}{\frac{\Gamma(1/\alpha_L)}{\alpha_L} + \frac{\Gamma(1/\alpha_H)}{\alpha_H}} \left(\mathbb{1}\{s \geq 0\} e^{-s^{\alpha_H}} + \mathbb{1}\{s \leq 0\} e^{-(-s)^{\alpha_L}} \right)$$

Recall that, by definition, $\frac{\Gamma(1/\alpha)}{\alpha}$ is increasing in α for all $\alpha \geq \alpha^*$. This means, in particular, that there is a larger mass of positive signals than negative signals in this case.

What must be shown is that

$$\frac{f(\tau)^2}{F(\tau)(1 - F(\tau))}$$

is maximized for a negative value of τ . For ease of notation, let $\alpha := \alpha_H$ and $\alpha' := \alpha_L$. Notice that as before, the numerator is strictly increasing in τ for $\tau < 0$, and strictly decreasing in τ for $\tau > 0$. Since $\alpha' > \alpha$, for small values of τ there is larger persistence in the value of f on the negative side of the distribution.¹⁶ Explicitly, given a $\tau > 0$, we will have that $-\tau^{\alpha/\alpha'} < -\tau$ (for $\tau < 1$, which will again be optimal here) will have the same value of the numerator. From here, the claim is two-fold.

Claim 1: The optimal choice of τ for $\tau > 0$ has $\tau \geq \tau^*(\alpha)$.

To see this, consider that the first order condition is given by

$$-2\alpha\tau^{\alpha-1} - e^{-x^\alpha} \left[\frac{1}{\frac{\Gamma(1/\alpha')}{\alpha'} + \int_0^\tau e^{-x^\alpha} dx} - \frac{1}{\int_\tau^\infty e^{-x^\alpha} dx} \right].$$

As $\frac{\Gamma(1/\alpha')}{\alpha'} > \frac{\Gamma(1/\alpha)}{\alpha}$, the first fraction is smaller, so that the positive component is larger. This means that for the first order condition to be satisfied for $\tau > 0$ we need $\tau > \tau^*(\alpha)$.

Claim 2: For all $\tau > \tau^*(\alpha)$, it is the case that $1 - F(\tau) > F(-\tau^{\alpha/\alpha'})$.

Notice that

$$1 - F(\tau) = \frac{1}{\frac{\Gamma(1/\alpha)}{\alpha} + \frac{\Gamma(1/\alpha')}{\alpha'}} \int_\tau^\infty e^{-x^\alpha} dx = \frac{1}{\frac{\Gamma(1/\alpha)}{\alpha} + \frac{\Gamma(1/\alpha')}{\alpha'}} \cdot \frac{\Gamma\left(\frac{1}{\alpha}, \tau^\alpha\right)}{\alpha}$$

and

$$F(-\tau^{\alpha/\alpha'}) = \frac{1}{\frac{\Gamma(1/\alpha)}{\alpha} + \frac{\Gamma(1/\alpha')}{\alpha'}} \int_{\tau^{\alpha/\alpha'}}^\infty e^{-x^{\alpha'}} dx = \frac{1}{\frac{\Gamma(1/\alpha)}{\alpha} + \frac{\Gamma(1/\alpha')}{\alpha'}} \cdot \frac{\Gamma\left(\frac{1}{\alpha'}, \tau^\alpha\right)}{\alpha'}$$

From here it is sufficient to show that $\frac{\Gamma(1/\alpha, \tau^*(\alpha)^\alpha)}{\alpha} > \frac{\Gamma(1/\alpha', \tau^*(\alpha)^\alpha)}{\alpha'}$ for all $\alpha' > \alpha$. Notice that the value of τ at which this is evaluated is important: it does not hold for all τ (indeed, it clearly will not hold for $\tau = 0$).

Subclaim: $\frac{\Gamma(1/\alpha, \tau^\alpha)}{\alpha} > \frac{\Gamma(1/\alpha', \tau^\alpha)}{\alpha'}$ for $\alpha \geq \alpha^*$, $\tau^\alpha \geq \tau^*(\alpha^*)^{\alpha^*} > 0.001$.

Now, do the standard change of variable for $\beta := 1/\alpha$ and $x := \tau^\alpha$. I show that

¹⁶Of course, for $\tau > 1$ this is reversed, because the tails are smaller on the negative side of the distribution than the positive side of the distribution.

for $\beta < 1/\alpha^*$ it is the case that

$$\frac{\partial}{\partial \beta} (\Gamma(\beta, x) \cdot \beta) > 0$$

for all $x > 0.001$. The above is equivalent to (after some straightforward algebra in the same vein as above)

$$-\frac{1}{\beta} < \frac{\int_x^\infty \log(y) d\lambda_\beta(y)}{\int_x^\infty d\lambda_\beta(y)} \quad (17)$$

where the $d\lambda_\beta(y)$ notation is the same as in Lemma 7. Notice first that the right-hand side of (17) is (i) increasing in x and (ii) increasing in β . Property (i) holds because $\log$ is increasing, and so by increasing x the average log value will increase. Property (ii) follows from a similar logic: smaller β put more weight on smaller values of y (and hence smaller values of $\log$).

This means that if (17) holds for $x^* := 0.001$, then the (sub)claim will be proved. From here, several straightforward computations will prove the claim:

$$-2.16 < \frac{\int_{x^*}^\infty \log(y) d\lambda_{1/2.7}(y)}{\int_{x^*}^\infty d\lambda_{1/2.7}(y)}, \quad (18)$$

dealing with $\beta \in (\frac{1}{2.7}, \frac{1}{\alpha^*}]$. Then,

$$-2.7 < \frac{\int_{x^*}^\infty \log(y) d\lambda_{1/4}(y)}{\int_{x^*}^\infty d\lambda_{1/4}(y)}, \quad (19)$$

dealing with then $\beta \in (\frac{1}{4}, \frac{1}{\alpha^*}]$. Finally, it is the case that

$$-4 < \frac{\int_{x^*}^\infty \log(y) d\lambda_0(y)}{\int_{x^*}^\infty d\lambda_0(y)}.$$

This means that for all $\beta \in (0, \alpha^*]$ it is the case that (17) holds. This proves Claim 2.

From here, the conclusion is reached by simply putting together Claims 1 and 2: for any possible optimal positive τ , there is a negative threshold that performs strictly

better, so the optimal threshold must be negative. $\square$

A.3 Proofs for Section 5

In this Appendix I provide the proofs of the results in Section 5.

Proof of Proposition 4. We show this in the case that r is a finite review system, and when γ^r has full support on R_r , since the notation is easier in this case. First, observe that $f_H(s) = f_L(s) \frac{1-\pi(s)}{\pi(s)}$. This means that we can write $\nu(r)$ as

$$\begin{aligned} 1 - \nu(r) &= \min_{\lambda \in [0,1]} \sum_{\tilde{r} \in R_r} \mathbb{P}_L(r^{-1}(\tilde{r}))^\lambda \mathbb{P}_H(r^{-1}(\tilde{r}))^{1-\lambda} \\ &= \min_{\lambda \in [0,1]} \sum_{\tilde{r} \in R_r} \left(\int_{\{s \in r^{-1}(\tilde{r})\}} \frac{\pi(s)}{(1-\pi(s))} f_H(s) ds \right)^\lambda \cdot \left(\int_{\{s \in r^{-1}(\tilde{r})\}} f_H(s) ds \right)^{1-\lambda} \end{aligned}$$

Now, if $\pi(s)$ is injective, it is the case (by straightforward computation) that $\frac{1}{1-\pi} f_H(s) = 2\gamma_\pi(d\pi(s))$ and so also $f_H(s) = 2(1-\pi(s))\gamma_\pi(d\pi(s))$. In general (even if $\pi(s)$ is not injective), for each λ , then,

$$\begin{aligned} &\sum_{\tilde{r} \in R_r} \mathbb{P}_L(r^{-1}(\tilde{r}))^\lambda \mathbb{P}_H(r^{-1}(\tilde{r}))^{1-\lambda} \\ &= \sum_{\tilde{r} \in R_r} 2 \left(\int_{\tilde{\pi} \in r^{-1}(\tilde{r})} \tilde{\pi} \gamma_\pi(d\tilde{\pi}) \right)^\lambda \cdot \left(\int_{\tilde{\pi} \in r^{-1}(\tilde{r})} (1-\tilde{\pi}) \gamma_\pi(d\tilde{\pi}) \right)^{1-\lambda} \\ &= \sum_{\tilde{r} \in R_r} 2 \frac{\left(\int_{\tilde{\pi} \in r^{-1}(\tilde{r})} \tilde{\pi} \gamma_\pi(d\tilde{\pi}) \right)^\lambda \cdot \left(\int_{\tilde{\pi} \in r^{-1}(\tilde{r})} (1-\tilde{\pi}) \gamma_\pi(d\tilde{\pi}) \right)^{1-\lambda}}{\int_{\tilde{\pi} \in r^{-1}(\tilde{r})} \gamma_\pi(d\tilde{\pi})} \int_{\tilde{\pi} \in r^{-1}(\tilde{r})} \gamma_\pi(d\tilde{\pi}) \\ &= \sum_{\tilde{r} \in R_r} 2 \left(\mathbb{E}_{\gamma_\pi} [\pi | r(\pi) = \tilde{r}] \right)^\lambda \left(1 - \mathbb{E}_{\gamma_\pi} [\pi | r(\pi) = \tilde{r}] \right)^{1-\lambda} \int_{\tilde{\pi} \in r^{-1}(\tilde{r})} \gamma_\pi(d\tilde{\pi}) \end{aligned}$$

From here, we minimize over λ , and conclude. $\square$

A.3.1 Optimality of Threshold Rules

If r is a consistent review system that has b bins, then it is effectively a partition of the space $[0, 1]$. Let $\{B_1, \dots, B_b\}$ be the partition of the interval. Then (6) can be

rewritten as

$$\nu(r) = 1 - \min_{\lambda \in [0,1]} 2 \sum_{i=1}^b \left(\mathbb{E}_{\gamma_\pi} [\pi | \pi \in B_i] \right)^\lambda \left(1 - \mathbb{E}_{\gamma_\pi} [\pi | \pi \in B_i] \right)^{1-\lambda} \gamma_\pi(B_i).$$

With this structure, it is possible to show the intuitive result that optimal coarsening with a fixed number of bins must necessarily be threshold rules in posterior space. First, I explicitly define what it means to be defined by thresholds.

Definition 5. A review system r is called a *b-threshold* review function if $r^{-1}(\tilde{r})$ is convex (except perhaps on a set of measure zero) for each $\tilde{r} \in R_r$.

Optimal review systems are necessarily threshold rules.

Proposition 5. *Let r be an optimal review system with b bins. Then r is b-threshold review function.*

Proof of Proposition 5. I show that if r is a finite review with b bins that is not such a threshold rule, then there exists a review with b bins that outperforms it. Moreover, I show that this holds point-wise for each λ , so that certainly it holds over the minimum of λ .

Consider two bins r_1 and r_2 that cannot be separated by any threshold. Let $\pi_i = \mathbb{E}[\pi | r(\pi) = r_i]$. Without loss assume that $\pi_1 \leq \pi_2$.

Consider any point $\bar{\pi}$ that lies between π_1 and π_2 (if they are equal, then $\bar{\pi} = \pi_1 = \pi_2$). By assumption, $\bar{\pi}$ is not a threshold for these two bins. In particular, this means that because $\pi_1 \leq \bar{\pi} \leq \pi_2$, there exists a set Π_1 with positive measure such that for all $\pi \in \Pi_1$, $\pi > \bar{\pi}$ and $r(\pi) = r_1$. That is, consider the following two sets

$$\begin{aligned} \Pi_1 &:= \{\pi | r(\pi) = r_1 \text{ and } \pi > \bar{\pi}\} \quad \text{and} \\ \Pi_2 &:= \{\pi | r(\pi) = r_2 \text{ and } \pi < \bar{\pi}\}. \end{aligned}$$

Because $\bar{\pi}$ is not a threshold for r_1, r_2 , and $\pi_1 \leq \bar{\pi} \leq \pi_2$, these sets both must have positive measure. Suppose that both of their measures are larger than ϵ . Select a subset of each that has mass ϵ : $\Pi_1^\epsilon, \Pi_2^\epsilon$. Let $\pi'_i = \mathbb{E}[\pi | r'(\pi) = r_i]$ denote the new conditional expectations. Because of the construction it must necessarily be the case that $\pi_1 > \pi'_1$ and $\pi_2 < \pi'_2$.

Now, define a new review function r' as follows:

$$r'(\pi) = \begin{cases} r(\pi) & \text{if } \pi \notin \Pi_1^\epsilon, \Pi_2^\epsilon \\ r_1 & \text{if } \pi \in \Pi_2^\epsilon \\ r_2 & \text{if } \pi \in \Pi_1^\epsilon \end{cases}.$$

That is, r' agrees with r , except that it swaps the Π_i^ϵ 's. I show that we necessarily have $\nu(r') \geq \nu(r)$. As mentioned, I show that this holds for each λ . To that end, fix a value of $\lambda \in (0, 1)$, and let $\varphi(\pi) := \pi^\lambda(1 - \pi)^{1-\lambda}$. I now show that the iterated expectation of $\varphi(\pi)$ on the partition induced by r' is lower than on the partition induced by r .

That is, I aim to show that

$$\sum_{i=1}^b \varphi\left(\mathbb{E}_{\gamma_\pi}[\pi | r(\pi) \in r_i]\right) \gamma_\pi(r^{-1}(r_i)) \geq \sum_{i=1}^b \varphi\left(\mathbb{E}_{\gamma_\pi}[\pi | r'(\pi) \in r_i]\right) \gamma_\pi((r')^{-1}(r_i)).$$

Notice now that the two agree except on r_1 and r_2 , so that the sum simplifies. Now, consider adding a linear function $h(\pi)$ to $\varphi(\pi)$ such that the unique maximum of $h + \varphi$ is obtained at $\bar{\pi}$. Note that this is possible because φ is strictly concave. In particular, this means that $h + \varphi$ is strictly increasing for $\pi < \bar{\pi}$ and strictly decreasing for $\pi > \bar{\pi}$. Then, because h is linear, it is the case that

$$\begin{aligned} \sum_{i=1}^2 \varphi(\pi_i) \gamma_\pi(r^{-1}(r_i)) &\geq \sum_{i=1}^2 \varphi(\pi'_i) \gamma_\pi((r')^{-1}(r_i)) \\ \Leftrightarrow \sum_{i=1}^2 (\varphi(\pi_i) + h(\pi_i)) \gamma_\pi(r^{-1}(r_i)) &\geq \sum_{i=1}^2 (\varphi(\pi'_i) + h(\pi'_i)) \gamma_\pi((r')^{-1}(r_i)). \end{aligned}$$

As observed, it is the case that $\bar{\pi} \geq \pi_1 > \pi'_1$. This means that

$$\varphi(\pi_1) + h(\pi_1) > \varphi(\pi'_1) + h(\pi'_1)$$

It is similarly the case that

$$\varphi(\pi_2) + h(\pi_2) > \varphi(\pi'_2) + h(\pi'_2)$$

Since by construction it is the case that $\gamma_\pi(r^{-1}(r_i)) = \gamma_\pi((r')^{-1}(r_i))$ for $i = 1, 2$, this means that we have achieved a strict improvement. Hence it must be the case that $\nu(r) < \nu(r')$. $\square$

A.4 Proofs for Section 6

In this Appendix I provide the proofs of the results in Section 6.

A.4.1 Proofs for Section 6.1

Micro-founding Separable Propensities: I suppose now that reviewers are defined by three characteristics. As before, reviewer i receives a conditionally independent signal s_i about the state of the world. Now, however, their willingness to review $w_i = (w_{i,1}, w_{i,2})$ is comprised of two components.

As before, I assume that s_i and w_i are independent, and also that $w_{i,1}$ and $w_{i,2}$ are independent. The reviewer's decision to review now follows a two-stage procedure. First, as before, the reviewer compares $w_{i,1}$ to a *review-dependent* threshold, which captures the heterogeneity in reporting rates across reviews. Let p_r be the probability that a reviewer "passes" this first stage. This proportion p_r can be thought of as the proportion of the population who would ever leave a review of type r : many consumers never leave written reviews, but are open to leaving a five-star review.

Second, the reviewer compares $w_{i,2}$ to a *signal-dependent* threshold. Let $q(s)$ be the probability with which a reviewer with signal s passes this second stage, conditional on passing the first stage, and assume that $q(s)$ is continuous as a function of s . That is, within the population of mass p_r who might leave a review of type r , among those with signal s only proportion $q(s)$ will actually leave the review: not everyone who sometimes writes reviews will always leave a review. Together, these factors combine to mean that the probability that a reviewer with signal s facing review r leaves a review is $p_r q(s)$.

Proof of Theorem 3. I proceed with the computations in many steps. At risk of abusing notation, let

$$f(s, \mu, \lambda) := f(s + \mu)^\lambda f(s - \mu)^{1-\lambda}$$

For an abuse of notation, let $p(s) := p(r, s) = p_r q(s)$ when the review function in question is clear. let $\rho(\mu; p, \lambda)$ denote the value of $\rho(\mu; p)$ evaluated at a fixed (but not necessarily the optimal) λ .

It can easily be seen that $\rho(\mu; p), \rho(\mu; r, p) \rightarrow 0$. Hence, l'Hopital's rule must be applied. First, we look at $\lim_{\mu \rightarrow 0} \frac{\partial}{\partial \mu} \rho(\mu; p)$. The regularity assumptions on f mean that, for a fixed λ , we have that this is given by

$$\begin{aligned} -\frac{\partial}{\partial \mu} \rho(\mu; p, \lambda) = & \int \left(\lambda \frac{f'(s + \mu)}{f(s + \mu)} - (1 - \lambda) \frac{f'(s - \mu)}{f(s - \mu)} \right) f(s, \mu, \lambda) p(s) ds \\ & + \lambda \frac{\int ((1 - p(s)) f(s + \mu) ds)^{\lambda-1}}{\int ((1 - p(s)) f(s - \mu) ds)^{\lambda-1}} \int ((1 - p(s)) f'(s + \mu) ds) \\ & - (1 - \lambda) \frac{\int ((1 - p(s)) f(s + \mu) ds)^{\lambda}}{\int ((1 - p(s)) f(s - \mu) ds)^{\lambda}} \int ((1 - p(s)) f'(s - \mu) ds). \end{aligned}$$

Notice that because of the envelope theorem, we need not consider here the derivative with respect to λ . Consider now the limit as $\mu \rightarrow 0$. First, $f(s, \mu, \lambda) \rightarrow f(s, \mu, \lambda)$. Hence, the limit is equal to

$$-\lim_{\mu \rightarrow 0} \frac{\partial}{\partial \mu} \rho(\mu; p, \lambda) = (2\lambda - 1) \int p(s) \ell_f(s) f(s) ds + (2\lambda - 1) \int (1 - p(s)) f'(s) ds.$$

This is equal to zero because $\ell_f(s) f(s) = f'(s)$, and $\int f'(s) ds = 0$ (because the expectation of the linear score function is zero).

Hence, we must consider the second derivative. Here it can be verified that the derivative with respect to λ will indeed be irrelevant, because this value is zero. This means that we need not consider how λ varies with μ , and again may consider evaluating at λ point-wise.

Because the notation gets so tedious, for readability of the null review terms, let $H_+(\mu, n) := \int (1 - p(s)) f(s + \mu) ds$ (where n refers to “null”), and similarly $H_-(\mu, n)$. In a similar vein, let $H'_+(\mu, n) := \int (1 - p(s)) f'(s + \mu) ds$, and similarly $H'_-(\mu, n)$ (and $H''_{\pm}(\mu, n)$). Computing, we get that

$$-\frac{\partial^2}{\partial \mu^2} \rho(\mu; p, \lambda) =$$

$$\begin{aligned}
& \int \lambda(\lambda - 1) \left(\left(\frac{f'(s + \mu)}{f(s + \mu)} \right)^2 + \left(\frac{f'(s + \mu)f'(s - \mu)}{f(s + \mu)f(s - \mu)} \right) \right) f(s, \mu, \lambda) p(s) ds \\
& - \int \lambda(1 - \lambda) \left(\left(\frac{f'(s - \mu)}{f(s - \mu)} \right)^2 + \left(\frac{f'(s - \mu)f'(s + \mu)}{f(s - \mu)f(s + \mu)} \right) \right) f(s, \mu, \lambda) p(s) ds \\
& + \int \left(\lambda \frac{f''(s + \mu)}{f(s + \mu)} + (1 - \lambda) \frac{f''(s - \mu)}{f(s - \mu)} \right) f(s, \mu, \lambda) p(s) ds \\
& + \lambda(\lambda - 1) \frac{H_+(\mu, n)^{\lambda-1}}{H_-(\mu, n)^{\lambda-1}} \left(\frac{H'_+(\mu, n)}{H_+(\mu, n)} + \frac{H'_-(\mu, n)}{H_-(\mu, n)} \right) H'_+(\mu, n) \\
& - (1 - \lambda) \lambda \frac{H_+(\mu, n)^\lambda}{H_-(\mu, n)^\lambda} \left(\frac{H'_+(\mu, n)}{H_+(\mu, n)} + \frac{H'_-(\mu, n)}{H_-(\mu, n)} \right) H'_-(\mu, n) \\
& + \lambda \frac{H_+(\mu, n)^{\lambda-1}}{H_-(\mu, n)^{\lambda-1}} H''_+(\mu, n) + (1 - \lambda) \frac{H_+(\mu, n)^\lambda}{H_-(\mu, n)^\lambda} H''_-(\mu, n).
\end{aligned}$$

We can then see that the limit is given by

$$\begin{aligned}
& \lim_{\mu \rightarrow 0} \frac{\partial^2}{\partial \mu^2} \rho(\mu; p, \lambda) = \\
& 4\lambda(1 - \lambda) \int \left(\frac{f'(s)}{f(s)} \right)^2 f(s) p(s) + \int f''(s) p(s) ds \\
& + 4\lambda(1 - \lambda) \frac{H'_+(0, n)^2}{H_+(0, n)} + H''_+(0, n) \\
& = 4\lambda(1 - \lambda) \int \left(\frac{f'(s)}{f(s)} \right)^2 f(s) p(s) + \int p(s) f''(s) ds \\
& + 4\lambda(1 - \lambda) \frac{(\int (1 - p(s)) f'(s) ds)^2}{\int (1 - p(s)) f(s) ds} + \int (1 - p(s)) f''(s) ds.
\end{aligned}$$

Now, as $\int f''(s) ds = 0$ (because its anti-derivative is f'), $\int f'(s) ds = 0$, and $\int f(s) ds = 1$, this value is equal to the numerator in the statement of the proposition when $\lambda = \frac{1}{2}$. Notice that this value of λ is clearly the maximizer of this object. minimized object is the negative of this, $\lambda = \frac{1}{2}$ will minimize the numerator, so it is the optimal value of λ .

Of course, at this stage it has not been shown that the computations that were computed above correspond to the value of the numerator: the denominator needs to go to 0 at the same rate for this to be satisfied. However, for linearity of the argument it is worthwhile to perform all of the above computations in order. I now

turn attention to the denominator. The terms in the denominator are of the following form. From hereon $p(s)$ to refer to $p_r q(s)$. At the end, we shall replace these values with their true values. Define

$$g_{\tilde{r}}(\mu, \lambda) := \left(\int_{\{s: r(s) \in \tilde{r}\}} f(s + \mu) p(s) ds \right)^\lambda \left(\int_{\{s: r(s) \in \tilde{r}\}} f(s - \mu) p(s) ds \right)^{1-\lambda}.$$

Similarly, for null reviews we have

$$g_n(\mu, \lambda) := \left(\int f(s + \mu)(1 - p(s)) ds \right)^\lambda \left(\int f(s - \mu)(1 - p(s)) ds \right)^{1-\lambda}.$$

The value of the denominator for a fixed λ is of the form

$$\rho(\mu; r, p, \lambda) := 1 - \sum_{\tilde{r} \in R_r} g_{\tilde{r}}(\mu, \lambda) - g_n(\mu, \lambda).$$

In order to make computations easier to follow, I compute the limits term by term. Notice that the computations for $g_n(\mu, \lambda)$ have already been computed above. For notational clarity, define

$$H_+(\mu, \tilde{r}) = \int_{\{r(s) \in \tilde{r}\}} f(s + \mu) p(s) ds.$$

and similarly all of the related functions $(H_-(\mu, \tilde{r}), H'_\pm(\mu, \tilde{r}), H''_\pm(\mu, \tilde{r}))$. Notice here that the term is $p(s)$ and not $1 - p(s)$, because these are submitted reviews. Let

$$\frac{\partial}{\partial \mu} g_{\tilde{r}}(\mu, \lambda) = \lambda \frac{H_+(\mu, \tilde{r})^{\lambda-1}}{H_-(\mu, \tilde{r})^{\lambda-1}} H'_+(\mu, \tilde{r}) - (1 - \lambda) \frac{H_+(\mu, \tilde{r})^\lambda}{H_-(\mu, \tilde{r})^\lambda} H'_-(\mu, \tilde{r}).$$

This means that

$$\lim_{\mu \rightarrow 0} \frac{\partial}{\partial \mu} g_{\tilde{r}}(\mu, \lambda) = (2\lambda - 1) H'(0, \tilde{r}).$$

Recall that we have already computed the dynamics of $g_n(\mu, \lambda)$ above. This means

that, regardless of λ , we have that

$$\begin{aligned}
-\lim_{\mu \rightarrow 0} \frac{\partial}{\partial \mu} \rho(\mu; r, p, \lambda) &= (2\lambda - 1) \left(\sum_{\tilde{r}} H'_+(0, \tilde{r}) + H'_+(0, n) \right) \\
&= (2\lambda - 1) \left(\sum_{\tilde{r}} \int_{\{r(s) \in \tilde{r}\}} f'(s) p(s) ds + \int f'(s) (1 - p(s)) ds \right) \\
&= (2\lambda - 1) \int f'(s) ds = 0.
\end{aligned}$$

The reasons that we need not consider λ are similar to the reasons in the case of the numerator. Because of this, I may focus on point-wise evaluation with respect to λ throughout. In particular, this means that we must proceed and take another derivative. We have that

$$\begin{aligned}
\frac{\partial^2}{\partial \mu^2} g_{\tilde{r}}(\mu, \lambda) &= \lambda(\lambda - 1) \frac{H_+(\mu, \tilde{r})^{\lambda-1}}{H_-(\mu, \tilde{r})^{\lambda-1}} \left(\frac{H'_+(\mu, \tilde{r})}{H_+(\mu, \tilde{r})} + \frac{H'_-(\mu, \tilde{r})}{H_-(\mu, \tilde{r})} \right) H'_+(\mu, \tilde{r}) \\
&\quad - (1 - \lambda) \lambda \frac{H_+(\mu, \tilde{r})^\lambda}{H_-(\mu, \tilde{r})^\lambda} \left(\frac{H'_+(\mu, \tilde{r})}{H_+(\mu, \tilde{r})} + \frac{H'_-(\mu, \tilde{r})}{H_-(\mu, \tilde{r})} \right) H'_-(\mu, \tilde{r}) \\
&\quad + \lambda \frac{H_+(\mu, \tilde{r})^{\lambda-1}}{H_-(\mu, \tilde{r})^{\lambda-1}} H''_+(\mu, \tilde{r}) + (1 - \lambda) \frac{H_+(\mu, \tilde{r})^\lambda}{H_-(\mu, \tilde{r})^\lambda} H''_-(\mu, \tilde{r}).
\end{aligned}$$

Notice that these computations are similar to those used for $g_n(\mu, \lambda)$ in the computation of the numerator's second derivative. Note again that we need not consider the variation in λ here.

This means that

$$\lim_{\mu \rightarrow 0} \frac{\partial^2}{\partial \mu^2} g_{\tilde{r}}(\mu, \lambda) = 4\lambda(\lambda - 1) \frac{H'_+(0, \tilde{r})^2}{H_+(0, \tilde{r})} + H''_+(0, \tilde{r}).$$

We are almost done. All that is left is to collect terms and notice again $\int f''(s) ds = 0$:

$$\begin{aligned}
\lim_{\mu \rightarrow 0} \frac{\partial^2}{\partial \mu^2} \rho(\mu; r, p, \lambda) &= - \sum_{\tilde{r}} \left(4\lambda(\lambda - 1) \frac{H'_+(0, \tilde{r})^2}{H_+(0, \tilde{r})} + H''_+(0, \tilde{r}) \right) \\
&\quad - 4\lambda(\lambda - 1) \frac{H'_+(0, n)^2}{H_+(0, n)} - H''_+(0, n)
\end{aligned}$$

$$=4\lambda(1-\lambda)\left(\sum_{\tilde{r}}\frac{H'_+(0,\tilde{r})^2}{H_+(0,\tilde{r})}+\frac{H'_+(0,n)^2}{H_+(0,n)}\right).$$

Putting this result together with the computations from the numerator above show the claim. $\square$

A.4.2 Proofs for Section 6.2

Proof of Proposition 2. The proof follows from an application to Theorem 3, in the case that $q \equiv 1$ and the review system r is a threshold system. To see this, observe that (with the notation of the proof of Theorem 3), if $\tilde{r}$ consists of those signals that fall between τ_{i-1} and τ_i , it is the case that

$$\begin{aligned} H'_+(0,\tilde{r})^2 &= \left(\int_{\tau_{i-1}}^{\tau_i} p_r f'(s) ds\right)^2 = p_r^2 (f(\tau_i) - f(\tau_{i-1}))^2 \quad \text{and} \\ H_+(0,\tilde{r}) &= \left(\int_{\tau_{i-1}}^{\tau_i} p_r f(s) ds\right) = p_r (F(\tau_i) - F(\tau_{i-1})). \end{aligned}$$

The result follows from simply taking the ratio. $\square$

B Relationship between the Platform and Consumers' Preference over Information Structures

In this section, I study the perspective of a consumer choosing which source of information to use when making their decision. In practice, many platforms elicit multiple types of information from reviewers; often one can leave a coarse rating and then supplement the rating with a free-text full review.

In the work above I take the perspective of the platform *ex ante* choosing what type of review to elicit. In that choice the platform internalizes the randomness in the review decisions of reviewers. If there are multiple types of reviews that have been collected, when it comes time for consumers to choose between them, this randomness has been removed. So, the comparison is not between the rates of acquisition, but directly between the number of reviews of each type of review.

The comparison from the side of consumers is more similar to asking how many coarsened reviews “equals” one review of another type. The distinction between the

two is subtle, but as shown below, the removal of risk leads the consumer to require more coarsened reviews. However, as individual reviews become uninformative, this difference disappears and so the metric of learning loss from Theorem 2 applies again.

Lemma 8. *Fix a consumer with a (finite) decision problem, two review systems r, r' , and a degree of distinction μ . Then there exists an N large such that for all $n_r, n_{r'} > N$, n_r reviews of type r is preferred to $n_{r'}$ reviews of type r' if*

$$\frac{n_r}{n_{r'}} > \frac{\log(1 - \nu(\mu; r'))}{\log(1 - \nu(\mu; r))}.$$

Proof. Let (n_r, r) denote the statistical experiment that generates n_r conditionally independent signals of drawn according to r . Then, because each of the signals are independent,

$$\begin{aligned} \nu(\mu; (n_r, r)) &= 1 - \min_{\lambda \in [0, 1]} \int_R \dots \int_R \prod_{i=1}^{n_r} (\gamma_L^r(d\tilde{r}_i))^\lambda (\gamma_H^r(d\tilde{r}_i))^{1-\lambda} \\ &= 1 - \min_{\lambda \in [0, 1]} \prod_{i=1}^{n_r} \int_R (\gamma_L^r(d\tilde{r}_i))^\lambda (\gamma_H^r(d\tilde{r}_i))^{1-\lambda} \\ &= 1 - (1 - \nu(\mu; r))^{n_r}. \end{aligned}$$

From here, the result comes from an application of either Torgersen's theorem or Theorem 1. Note that this result follows the same logic as Proposition 3 in Mu et al. (2021). See also Chernoff (1952), which discusses this relationship at some length. $\square$

This condition is similar to, but not exactly, the condition of Theorem 1. As the next Proposition highlights, because the information for consumers is not random, consumers place more weight on “good” information than the platform. However, as information becomes very noisy, this difference shrinks and eventually disappears in the limit. Specifically, the relationship is as follows.

Proposition 6. *Let $\kappa(\mu; r', r) := \frac{\nu(\mu; r')}{\nu(\mu; r)}$ and let $\zeta(\mu; r', r) := \frac{\log(1 - \nu(\mu; r'))}{\log(1 - \nu(\mu; r))}$. Then,*

$$\kappa(\mu; r', r) < \zeta(\mu; r', r) \iff \nu(\mu; r') > \nu(\mu; r).$$

However,

$$\lim_{\mu \rightarrow 0} \kappa(\mu; r', r) = \lim_{\mu \rightarrow 0} \zeta(\mu; r', r).$$

Proof. For all $x, y \in (0, 1)$,

$$\frac{\log(1-x)}{\log(1-y)} > \frac{x}{y} \iff x > y.$$

This shows the first claim. For the second, one can apply l'Hopital's rule to the limit of $\mu \rightarrow 0$ for

$$\zeta(\mu; r', r) = \frac{\log(1 - \nu(\mu; r'))}{\log(1 - \nu(\mu; r))}.$$

and find that

$$\lim_{\mu \rightarrow 0} \zeta(\mu; r', r) = \lim_{\mu \rightarrow 0} \frac{\frac{\partial}{\partial \mu} \nu(\mu; r')}{\frac{\partial}{\partial \mu} \nu(\mu; r)} = \lim_{\mu \rightarrow 0} \eta(\mu; r', r).$$

This shows the second claim, completing the proof. □

The platform's choice between ratings systems must necessarily reflect the randomness with which it receives information. This means that, relative to a consumer who does not need to factor-in this randomness, the platform will value more frequent, but uninformative, reviews. This result suggests why in practice platforms collect multiple sources of information: consumers trade-off between sources of information in different ways than do platforms, because of the timing of their choice between sources of information.